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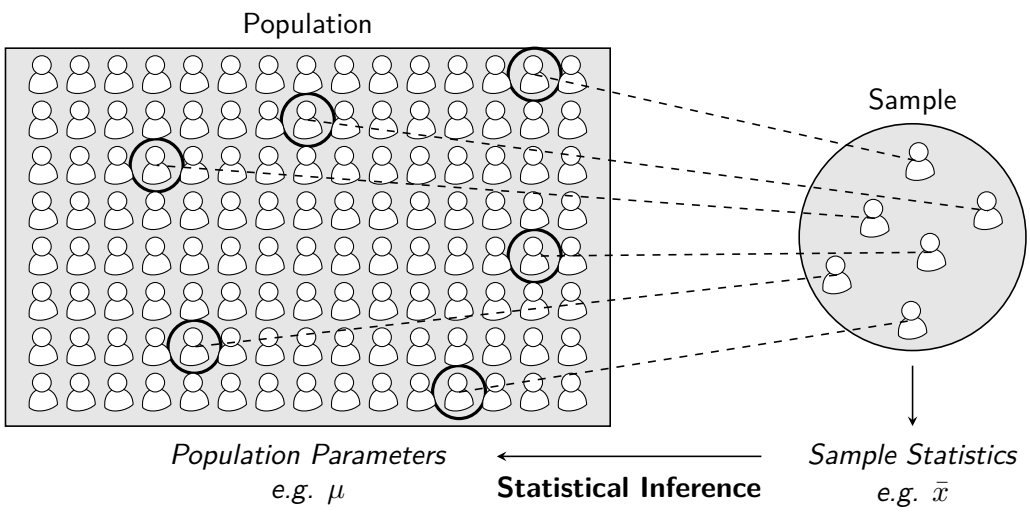
Introduction

Statistics is the science of collecting, organising, presenting and analysing data. It is vital in helping us develop a better understanding of the world, and to allow decisions to be made in a rational manner based on the information that is available to us. Governments, organisations, businesses and individuals across a diverse range of fields rely on statistics and statisticians.

The *Advanced Higher Statistics* course aims to equip learners with a sound foundation of statistical knowledge. This textbook has been created to serve as a source of key facts, examples, and exercises to support progress through the course.

Population Parameters and Sample Statistics



The diagram above gives a quick overview of the typical role of a statistician wishing to answer questions related to a *population* of interest. This is often in the form of wishing to know more about some *population parameter* such as a population mean, μ. Since it is generally impractical to take a *census* of the entire population, a *sample* is instead obtained from that population, from which *sample statistics* may be calculated, such as the sample mean, $\bar{x}$. *Statistical inference* is the process by which claims about population parameters can be tested using sample statistics.

This course, and textbook, will cover all of the key stages involved in statistical research and investigations, including:

- Gathering data using appropriate *sampling techniques*.
- Initial exploration of data using *descriptive statistics*, including:
  - Visualising data with charts and graphs.
  - Calculating sample statistics.
- Using *statistical inference* to assess the evidence for claims about population parameters, including:
  - An introduction to *probability theory*, which underpins all statistical inference
  - Conducting hypothesis tests

- Constructing confidence intervals

Techniques suitable for a range of contexts will be covered, and it is expected that by the end of the course students should have sufficient knowledge and confidence to collect their own data and conduct their own statistical investigations.

Suggested Use Of The Book

Other Resources

Acknowledgements

1

Exploratory Data Analysis

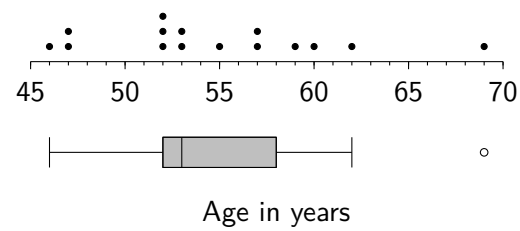
Datasets usually consist rows or columns of values, referred to as *raw data*. Before conducting any *statistical analysis*, a statistician will typically want to get a quick sense of what the data may reveal, as well as to identify any features of note. Common first steps are therefore often to: *organise* the data into suitable lists or tables; *visualise* the data using plots and graphs; and *measure* various properties of the data.

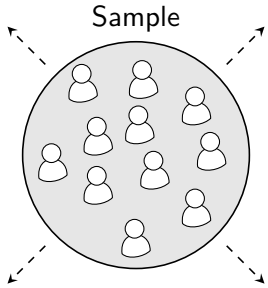
This process is often referred to as *exploratory data analysis*.

Patient data table (first five entries only)

age (yrs)	height (cm)	NHS health board
57	182.2	Fife
43	168.2	Tayside
48	179.4	Tayside
64	162.6	Forth Valley
52	171.5	Fife

Dot plot and box plot (age, NHS Fife patients)

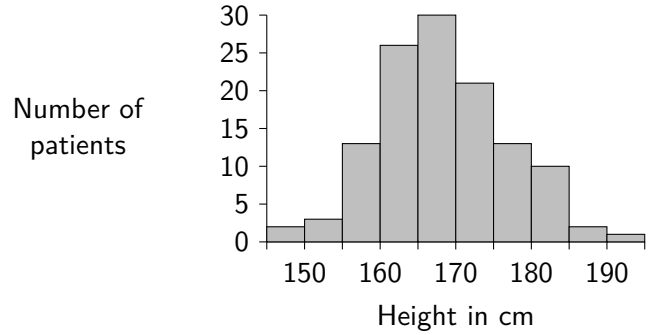




Summary (height in cm, all patients)

sample size	$n = 120$
mean	$\bar{x} = 168.5$
standard deviation	$s = 8.26$
median	med $= 168.9$
interquartile range	IQR $= 10.3$
minimum value	min $= 149.0$
maximum value	max $= 190.1$

Histogram (height, all patients)

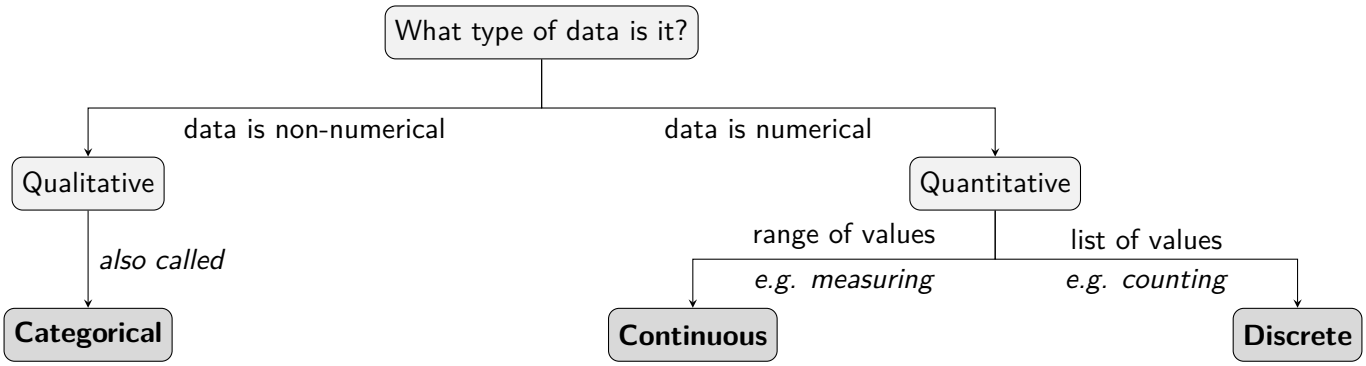


1.1 Types of Data

Knowing what *type* of data has been collected helps inform what kind of graphs and statistics may be appropriate.

Data that is numerical is called *quantitative*, of which there are two types. **Discrete** data consists of values that can be listed, such as someone's shoe size or the number of siblings they have. **Continuous** data can take *any* value within a range, though is often rounded to reflect the precision of measurements taken, such as the mass of a mouse.

Qualitative data (also called **categorical**) is *non-numerical*, such as whether a squirrel is red or grey.



For example, the table below shows data relating to some football players:

Position	Team	Height (m)	Caps	Goals
Midfielder	Aston Villa	1.78	66	18
Midfielder	Man United	1.93	49	8
Forward	Southampton	1.75	30	6
Forward	Bournemouth	1.78	49	6
Midfielder	Celtic	1.75	39	5
Categorical		**Continuous**	**Discrete**	

Exercise 1.1

1. A survey is carried out on the pets owned by a class. State whether the following data is qualitative or quantitative.

   (a) How many pets each pupil owns. (b) What kind of animal the pets are.

   (a) The pet's age. (b) The name of the pet.

   (a) The mass of the pet. (b) Whether the pet is a mammal.

2. For each of the following, state whether the data collected would be discrete or continuous.

   (a) The number of people in a room. (b) The mass of a book.

   (a) The number of pages in a book. (b) The number of songs in an album.

   (a) The volume of water in a bottle. (b) Time taken to complete a puzzle.

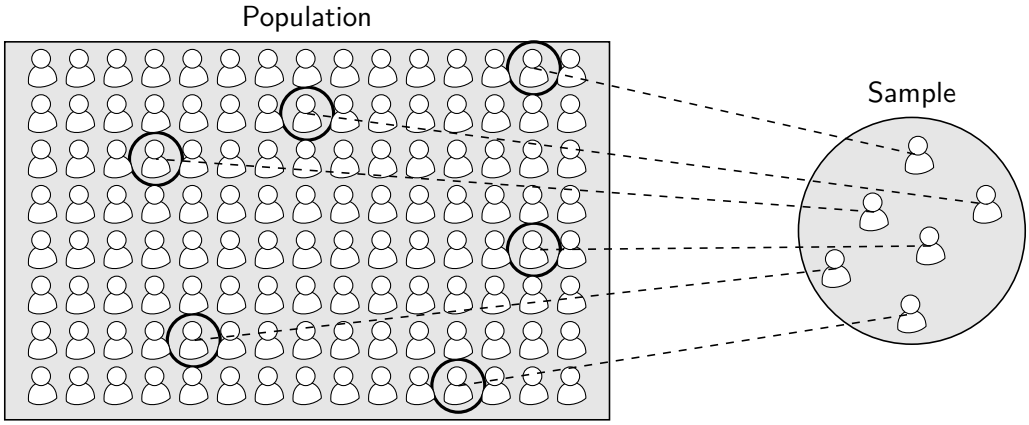
3. A random sample of members of a gym were surveyed and the first few results are shown in the table below.

Activity level	Smoker	Height (inches)	Mass (pounds)	Pulse (bpm)
Moderate	No	66	140	64
Moderate	No	72	145	58
A lot	Yes	73.5	160	62
Slight	Yes	73	190	66

   State the type of data that was collected for each variable.

1.2 Population Parameters and Sample Statistics

In a statistical investigation, the term **population** refers to the entire set of objects of interest, whilst a **sample** is a selection of some objects chosen from the population. Recognising the difference is fundamental to statistical analysis.

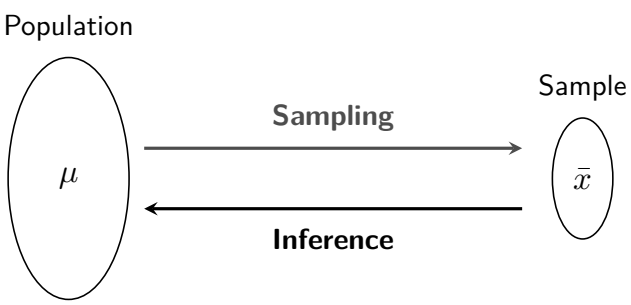


If data were available for an entire *population* then various **population parameters** could be calculated, such as the *population mean*. Since this is never really possible, instead *samples* are taken from populations. Measures calculated from a sample of data are called **sample statistics**, such as the *sample mean*.

Letters from the Greek alphabet are commonly used to represent population parameters, such as:

Population Parameter	**Sample Statistic**
Population mean μ *("mu")*	Sample mean $\bar{x}$ *("x bar")*
Population standard deviation σ *("sigma")*	Sample standard deviation s

It is good practice to refer *clearly* to whether a population parameter or sample statistic is being discussed.



Population parameters are seen as having fixed values, at the time of the study at least, whilst the values of sample statistics will depend on the sample chosen.

Sample statistics may be seen as *estimates* of the true value of their respective population parameters, and **inference** is the process by which claims may be made about population parameters (such as μ)based on observed sample statistics (such as $\bar{x}$).

Exercise 1.2

For each, state using the context of the question which population parameter or sample statistic has been obtained.

1. A year group at a school contains 120 pupils. Data is obtained for the number of merits each pupil in the year group obtained, and the mean of those values is calculated to be 3.4.
2. A bus company in Scotland checks the total mileage for some of its buses when they come in for a service. The mean mileage is calculated as 45,300 miles.
3. A medical trial for a new supplement to increase iron levels in the blood involves a number of patients being given the treatment over the course of a month, and then their blood iron levels being checked. The mean increase for the patients is found to be 27.3 micrograms per decilitre (mcg/dL).

1.3 Measures of Location and Spread

Summary statistics are a collection of measures taken of the data from a sample, such as the mean, or standard deviation. These offer a quick way to communicate some general sense of the data. For example, for the *raw data* of $0, 2, 2, 6, 12, 14, 16, 18, 18, 18, 20$, the table below shows a selection of summary statistics.

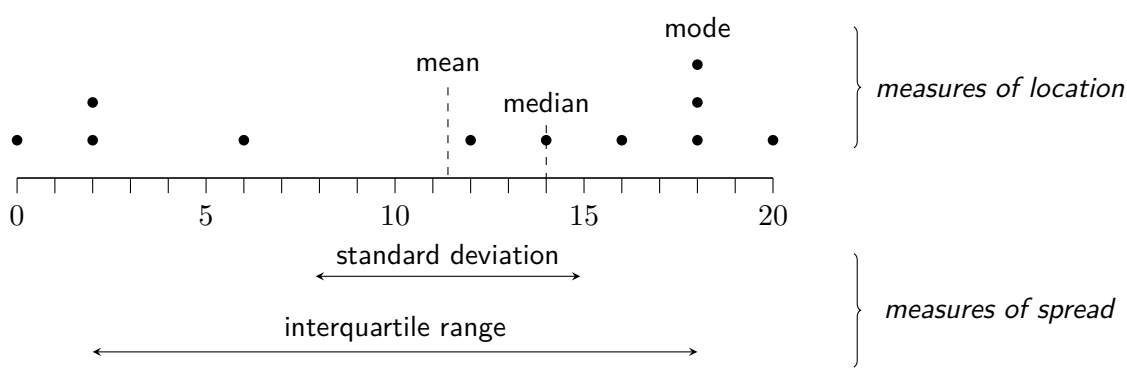
n	mean	sd	sum	sum of squares	min	LQ	median	UQ	max	range	IQR	mode
11	11.45	7.54	126	2012	0	4	14	18	20	20	14	18

There are two main types of summary statistic: those that measure the *central tendency* of data (or its **location**); and those that measure the *variability* of data (or its **spread**). It is common to wish to measure both the location *and* spread.

Measures of Location

The three main measures of location, or *averages*, should already be familiar. To avoid ambiguity, statisticians will typically avoid referring to *"the average"*, and will instead specify exactly *which* average they are referring to:

- **Mode** - the most commonly observed piece of data.
- **Median** - the middle value after the data has been put into numerical order.
- **Mean** - the sum of the values divided by the number of values.



Measures of Spread

The measure of spread which is usually encountered first in a maths classroom is the *range*, which is the difference between the highest and lowest values. In practice this measure is seldom used by statisticians since it only uses the two most extreme values from the data. There are three main measures of spread commonly used, two of which are likely to already be familiar:

- **Interquartile Range** - the difference between the upper quartile and the lower quartile.
- **Variance** - the *mean squared difference* between the values and the mean value.
- **Standard deviation** - the square root of the variance.

The Mean and the Standard Deviation

The *mean* and *standard deviation* are perhaps the most commonly used measures of location and spread, respectively, and both link strongly to the commonly-encountered *bell-shaped curve* (see page 13). The formulae for calculating the sample mean and sample standard deviation from sample data are given below using notation, where the Greek letter Σ (*"Sigma"*) means a *sum* should be calculated, x represents the all of the values in the sample (a shortened abbreviation of $x_1, x_2, \ldots, x_n$ or x_i) and n represents the *number* of values in the sample.

The *sample mean*, $\bar{x}$, is calculated as: $\bar{x} = \dfrac{\Sigma x}{n}$

The *sample standard deviation*, s, is calculated as: $s = \sqrt{\dfrac{S_{xx}}{n-1}}$ where: $S_{xx} = \Sigma x^2 - \dfrac{(\Sigma x)^2}{n}$

The *sample variance*, s^2, is simply the *square of the standard deviation*.

The Median and the Interquartile Range

The median, Q_2, and interquartile range, IQR, are often chosen to measure location and spread when the data is particularly *skewed* (see page 13), as they are less influenced by *extreme values* than the mean and standard deviation.

The *lower quartile* (LQ or Q_1) and the *upper quartile* (UQ or Q_3) split a set of values into quarters along with the median. From the lowest value to Q_1 contains the lowest 25% of the values, from Q_1 to Q_2 contains the next 25%, and so on.

Min $= 4$ $\quad$ $\mathrm{Q}_1 = 8$ $\quad$ $\mathrm{Q}_2 = 15$ $\quad$ $\mathrm{Q}_3 = 22$ $\quad$ Max $= 26$

$$4 \quad 5 \quad 8 \quad 8 \quad 10 \quad 11 \quad 13 \;\Big|\; 17 \quad 17 \quad 18 \quad 21 \quad 22 \quad 22 \quad 26$$

The *interquartile range* is calculated as: $\mathrm{IQR} = \mathrm{Q}_3 - \mathrm{Q}_1$

Exercise 1.3

1. The monthly cost of 10 drivers' car insurance (x, in £'s) are recorded. Calculate the mean and standard deviation.

$$\Sigma x = 359 \quad \Sigma x^2 = 15119 \quad n = 10$$

2. Daily fuel spend (f, in £'s) for 15 drivers is summarised below. Calculate the mean and standard deviation.

$$\Sigma f = 87 \quad \Sigma f^2 = 569 \quad n = 15$$

3. The marks, out of 40, of eight pupils sitting a chemistry test are below. Find the mean and standard deviation.

32 25 29 37 16 39 23 32

4. The ages (in years) of 18 customers in a shop are recorded. Calculate the median and interquartile range.

17	63	34	23	40	57	72	70	43
56	29	37	41	50	13	24	30	58

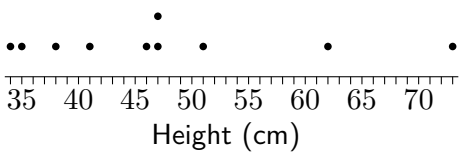
1.4 Displaying Data

Before any analysis of a data set is undertaken, it is common practice to *display* the data in a simple form so that an initial impression of what the data shows can be formed. Some key examples are shown below, using a sample taken by a gardener of the heights of 10 of their sunflowers 8 weeks after germination, measured to the nearest centimetre:

34 35 38 41 46 47 47 51 62 73

Dot Plots

Dot plots offer a quick way of presenting data in a manner similar to a bar chart, stacking repeated values.



Stem-and-Leaf Diagrams

Stem-and-leaf diagrams display data in a manner similar to a grouped bar graph. Leaves should be in order, and a key (or *legend*) and the sample size should be included.

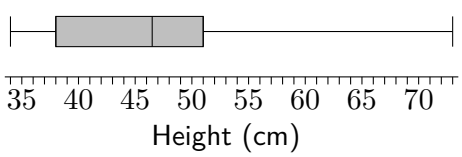
3	4 5 8
4	1 6 7 7
5	1
6	2
7	3

$n = 10$

3 | 5 means 35 cm

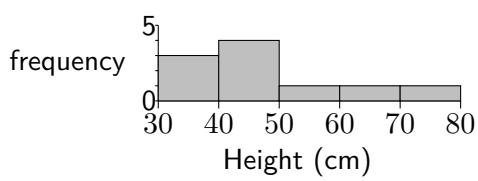
Box Plots

Box plots offer a visual representation of a five-figure summary, consisting of the mininum value, the lower quartile Q_1, the median, the upper quartile Q_3 and the maximum value.



Histograms

Histograms put continuous numerical data into intervals (or *bins*) and display it in a manner similar to a bar graph.



Exercise 1.4

1. The ages (in years) of competitors in a junior golf tournament are below. Draw a dot plot to represent this data.

   16 14 19 13 14 17 16 19 16 17 12 16 14 19

2. The price of an ice cream is recorded at 12 stalls, in £'s. Represent the data using a stem-and-leaf diagram.

   1.50 2.20 2.50 2.10 1.80 1.40
   2.00 1.90 3.60 0.90 1.70 1.50

3. The masses (in kg) of the luggage of 10 passengers on a plane to Barra were recorded. Construct a box plot.

   15 18 14 22 19 14 10 21 17 16

1.5 Tables, Figures and Output

There are many ways in which data can be presented, extending beyond those covered on the previous page. In statistical reports they are often clearly labelled as **tables** or **figures**, and numbered so they can be more easily referred to.

Table 1: Contingency table of mode of transport.

		Transport			
		walk	bus	car	bike
Year Group	3rd	12	15	3	4
	2nd	7	13	8	1
	1st	4	14	11	2

Figure 1: Bar chart of mode of transport.

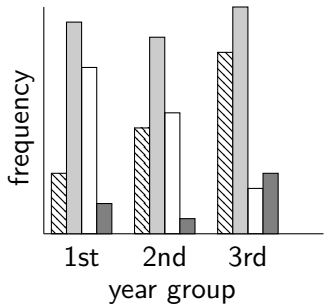
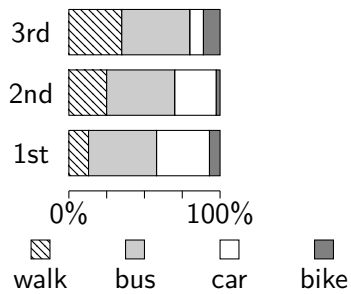


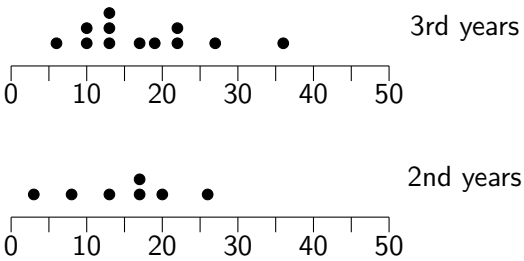
Figure 2: Chart of transport percentages.



Not all graphs encountered in real life will be up to scratch; some graphs may be unclear, accidentally misleading or even intentionally misleading. Being alert to potential flaws in tables and figures and identifying possible improvements is an important skill.

The widespread availability of computing power means that specialist software is often used by statisticians to avoid the need for time-consuming calculations where possible. Whilst this course requires no use of statistical software, examples of computer **output** will be displayed from which the required information must be extracted.

Figure 3: Dot plots of time (mins) to walk to school for 2nd and 3rd years.



Output 1: Summary data for walk times (mins) for 2nd and 3rd years.

```
data: walk time

3rd yr: n=12    mean=17.3   sd=8.42

2nd yr: n=**** mean=14.9   sd=7.65

alternative hypothesis: means are not equal

t=0.619, df=17, p-value=0.544
```

Whilst this textbook aims to enable familiarity with common figures, tables and computer output, the best way to practise obtaining information from them is to *read a wide range of relevant statistical articles and reports*.

Exercise 1.5

1. Use **Table 1** to calculate the proportion of pupils in the sample that walk to school.
2. Suggest two possible improvements to **Figure 1**.
3. Using **Figure 2**, compare the proportions of pupils in the sample who take the bus between each year group.
4. From **Figure 3**, state the approximate median time taken to walk for 3rd year pupils.
5. One piece of information from **Output 1** has been replace by ****. State the missing value.

1.6 Outliers and Fences

Sometimes a set of data may contain values that seem *extremely high or low* in comparison to the rest of the values. It is important to be wary of such values and the potentially significant impact they could have on any statistical analysis performed. There are a range of objective approaches that can be used to determine whether one or more points are extreme enough to be classed as **outliers**. In this course, *fences* will be used.

Identifying Outliers

$$\text{Lower Fence} = \text{Q}_1 - 1.5 \times \text{IQR}.$$

$$\text{Upper Fence} = \text{Q}_3 + 1.5 \times \text{IQR}.$$

Any values *below the lower fence* or *above the upper fence* are **outliers**.

If an outlier is identified in a set of data, an investigation should be conducted to determine the cause. If it is concluded that the outlier is a rogue point, such as an error in measurement, it can be removed from the data set and summary statistics re-calculated. However, if an outlier is a valid point in the data set then it should not be removed.

Returning to the sunflower heights from page 7, the data can now be checked for possible outliers:

Raw data

Heights (cm)

34	35	38	41	46	47	47	51	62	73

Summary data

n	min	Q1	median	Q3	max
10	34	38	46.5	51	73

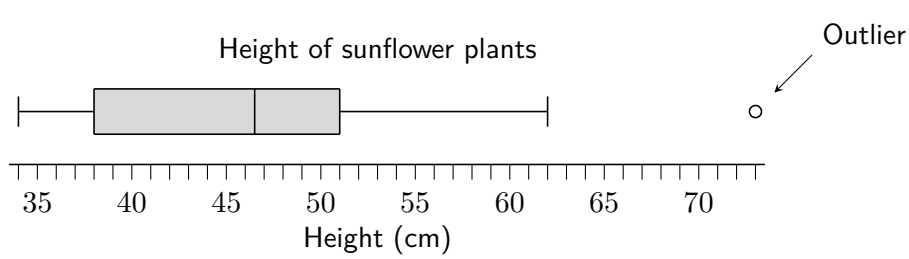
$$\text{IQR} = \text{Q}_3 - \text{Q}_1 = 51 - 38 = 13$$

$$\text{Lower fence} = \text{Q}_1 - 1.5 \times \text{IQR} = 38 - 1.5 \times 13 = 18.5$$

$$\text{Upper fence} = \text{Q}_3 + 1.5 \times \text{IQR} = 51 + 1.5 \times 13 = 70.5$$

Since $73 > 70.5$, it is an outlier.

On a box plot, outliers are indicated separately as shown below, with the "whiskers" extending only as far as the highest (and lowest) values which are *not* outliers. However the quartiles remain as those calculated using the *full* set of values, including any outliers. *Plotting outliers separately in this way is to highlight their presence, and is not a judgement on whether those values are rogue or valid.*



The outlier in this data set could be valid if, for instance, the sunflower plant is in a particular sunny spot in the garden. However, if the height was measured incorrectly then it can be removed from the data set.

Example

Problem: A vet conducts a survey, asking 10 randomly chosen cat owners registered with his practice to share the mass of their adult cat, in kilograms. The results are:

4.6 3.6 4.1 4.5 3.7 3.9 4.0 0.3 3.9 4.8

(a) Show that the set of data contains an outlier.

(b) Suggest a possible cause for this outlier which would mean it should be removed.

Solution:

(a) $Q_1 = 3.75, \quad Q_3 = 4.4, \quad \text{IQR} = 4.4 - 3.75 = 0.65.$

Lower fence $= 3.75 - 1.5 \times 0.65 = 2.775$

Upper fence $= 4.4 + 1.5 \times 0.65 = 5.375$

Since $0.3 < 2.775$, the value of 0.3 is an outlier.

(b) The owner could have mistakenly measured the mass of their cat in stones instead of kilograms.

Exercise 1.6

1. Strawberry tarts are sold every day in a bakery over the summer months. The shop manager monitors the sales and takes a random sample of the number of strawberry tarts sold on 10 randomly selected days over one month.

   23 31 45 32 35 19 24 25 30 38

   Calculate the upper and lower fences for this data set and identify any possible outliers.

2. The number of hours spent studying each week by a randomly selected group of 24 students is given below.

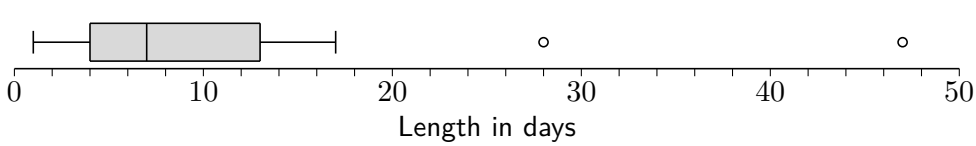
10	18	15	12	17	27	4	14	14	14	17	15
15	12	17	17	13	15	15	16	16	9	11	15

   (a) Identify any possible outliers.

   (b) Draw a box plot to display this data.

3. A hospital trust compiled data relating to the length of stay of patients in a particular hospital. A random sample of 18 patients gave the following summary data, measured in days.

n	mean	sd	min	Q1	median	Q3	max
18	10.78	11.17	1	4	7	13	47



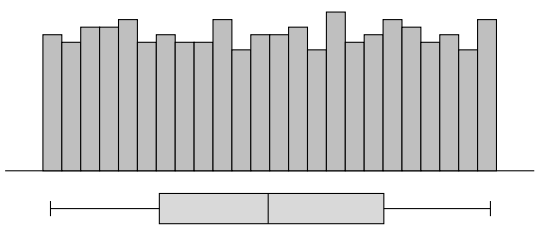
   (a) Show that the two values indicated by circles on the box plot are outliers.

   (b) Suggest a cause for these outliers that would make it valid to remove them from further analysis.

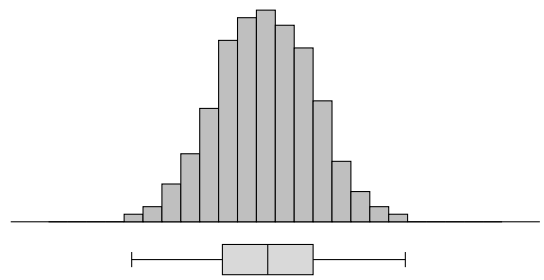
1.7 Describing Data Distributions

Summary statistics alone are not sufficient to describe the full complexity of the way data is distributed. It is also important to consider and be able to describe the **shape** of the distribution, accomplished most easily using a histogram. Whilst real-life rarely *perfectly* follows a precise mathematical model, a number of shapes are commonly observed and a number of descriptions are commonly considered. The following data was generated by *simulating sampling* from populations of data which followed the distributions described.

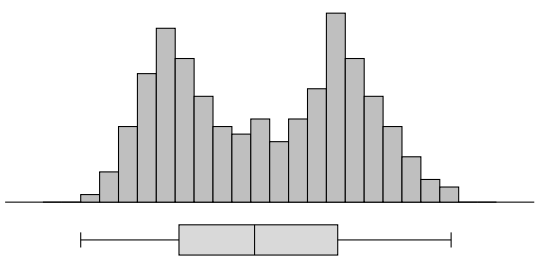
uniformly distributed data ($n = 103$)



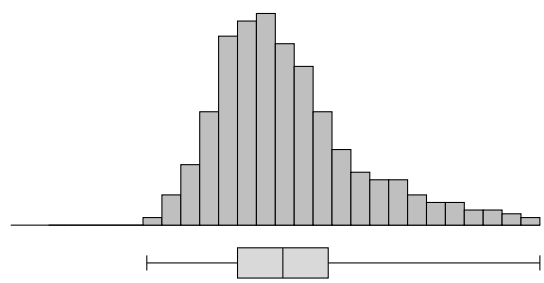
normally distributed data, also referred to as a bell-curve ($n = 215$)



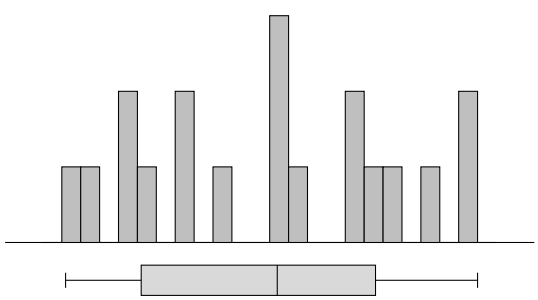
bimodal data ($n = 87$)



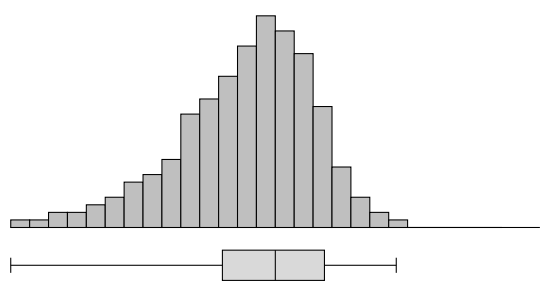
positively skewed data ($n = 134$)



very hard to infer distributions of data from small samples ($n = 19$)



negatively skewed data ($n = 97$)



As well as considering whether the **underlying population distribution** of data appears to be *normal*, *skewed*, *uniform* or *bimodal*, the appropriateness of some statistical inference techniques will depend on whether the distribution is *symmetrical*, or whether two sets of data follow the *same distribution*, with matching shapes and spread.

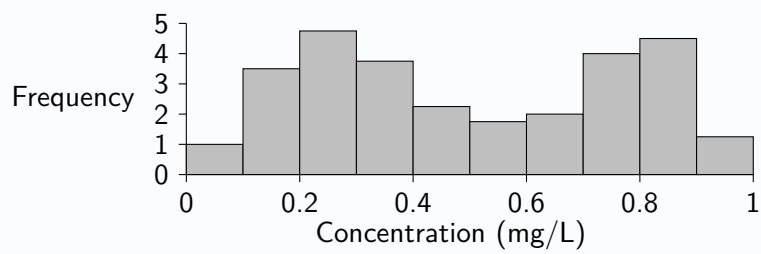
Whilst the following example references techniques that will be introduced later in the course, it is intended to give an idea of the importance of studying distributions in selecting appropriate methods for statistical inference.

Example

Problem: A environmental scientist collected water samples from a number of rivers in a region to explore whether the concentration of a harmful chemical in the water has changed since farming practises used locally have changed. They are considering the following choices of hypothesis test:

- A *t-test*, the most appropriate choice if the data follows a normal distribution.
- A *Wilcoxon Signed Rank test*, appropriate if the data is distributed symmetrically.

Figure 1: Concentration of chemical in river sites



(a) Use **Figure 1** to make a comment on the distribution of chemical concentrations.

(b) Comment on the appropriateness of each choice of hypothesis test suggested for this study.

Solution:

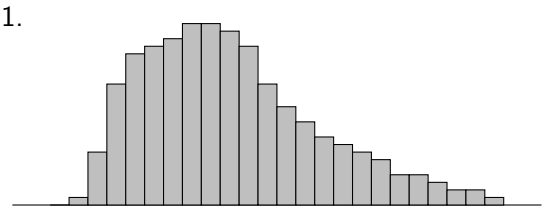
(a) Figure 1 suggests that the population distribution of concentrations of the chemical in river sites may be bimodal.

(b) Since the histogram suggests the population distribution of chemical concentration may be symmetrical but not normal, a Wilcoxon Signed Rank test would be appropriate whilst a t-test would not.

Exercise 1.7

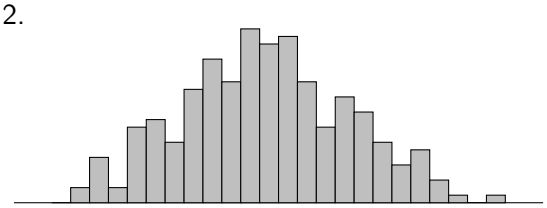
Comment on each population distribution using the simplified histograms, which show data from samples.

1.



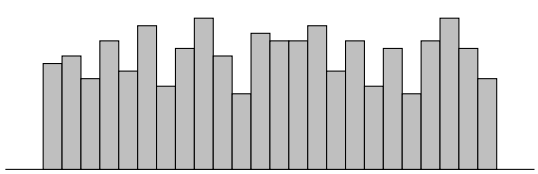
Hazelnut mass (grams)

2.



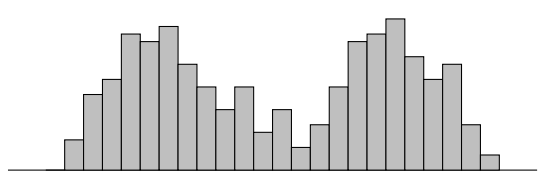
Tree height (metres)

3.



Waiting time (minutes)

4.



Number of successful passes

1.8 Comparing Distributions

When comparing two or more samples of data, a good starting point is to create suitable plots and obtain summary statistics. Differences and similarities should be noted, considering the location, spread and shape of the distributions of each group.

Example

Problem: Within the sport of professional cycling, it is common for riders to specialise in certain types of cycling. *Sprinters* aim to win races that finish on flat road at high speed, whilst *climbers* aim to win races with an uphill finish. A researcher studying the characteristics of elite-level cyclists randomly selects samples of 30 *sprinters* and 30 *climbers*, recording the heights of each. The results are shown summarised in **Table 1**.

Table 1

variable	n	Min	Q1	Median	Q3	Max	IQR
sprinters	30	165	174	179	184	200	10.5
climbers	30	158	168	174	177	195	9

Box plots for the two groups of cyclists are shown below in **Figure 1**, and dot plots in **Figure 2**.

Figure 1

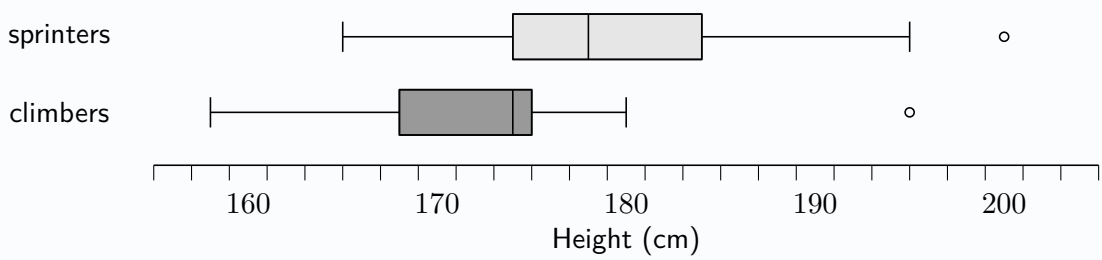
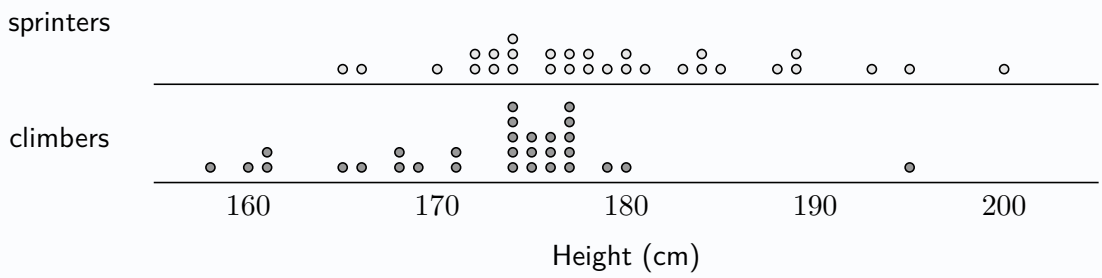


Figure 2



(a) Use **Figure 1** to make two comments comparing the heights of sprinters and climbers.

(b) Use **Figure 2** to comment on the shape of the distribution of the heights of climbers.

Solution:

(a) The sample median height of the sprinters was greater than that of the climbers.

The larger interquartile range suggests that the population heights of sprinters are more varied.

(b) The dot plot suggests the underlying distribution of heights for climbers is negatively skewed.

Note that this data is based on the heights of *samples* sprinters and climbers. Without the use of *statistical inference* techniques covered later in the course, it is only possible to *conjecture* rather than ever being able to make any definitive claims about the heights of the full *populations* of sprinters and climbers.

Exercise 1.8

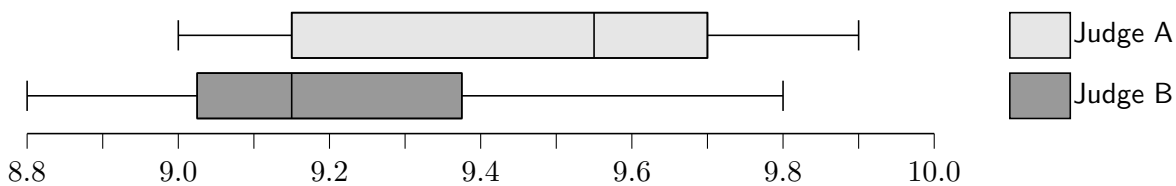
1. Two models of coffee machines produced by the brand *DOLCAFÉ* are supposed produce a white coffee with the same volume of liquid each time, but natural variation means the actual amount dispensed is not quite the same every time. The volumes dispensed, in ml, are recorded for random samples from each model.

Model A	295	298	299	301	303	304	309
Model B	282	294	300	302	306	312	320

   (a) Determine the sample mean and standard deviation for each model.

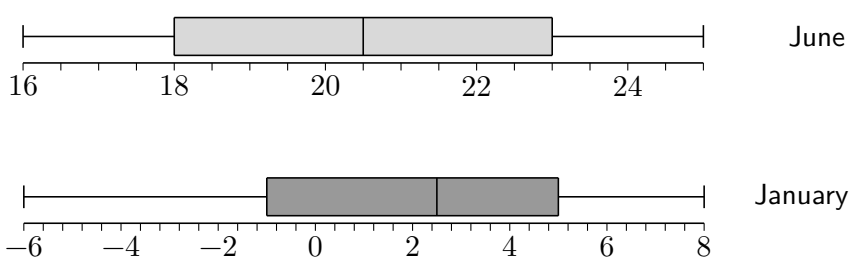
   (b) Make two comments comparing the amount of coffee dispensed by the two models.

2. The scores awarded by two judges in a diving competition are recorded and displayed in box plots, shown below.



   Using the box plots, make two comments comparing the scoring of the two judges.

3. The temperature at 2pm was noted for 14 randomly selected days in January and June at a local tourist spot and box plots drawn to display the results.



   (a) Using the box plots, make two comments comparing temperatures in January and June.

   (b) Explain why the box plots may be misleading.

4. The manager of a lemonade bottling plant is interested in comparing the performances of two production lines, one of which has only recently been installed. For each line she selects 10 one-hour periods at random and records the number of crates completed in each hour. The table below gives the results:

Production Line	Number of crates completed per hour									
New	78	87	79	82	87	81	85	80	82	83
Old	74	77	78	70	87	83	76	78	81	76

   (a) Draw dot plots to represent the data.

   (b) Calculate the mean and standard deviation for each sample.

   (c) Make two comments comparing the two lemonade production lines.

Review Exercise

1. A music fan wishes to investigate the characteristics of songs that are popular on a well-known music streaming app. She is able to obtain data on the thousand most-streamed songs of all time, a small selection of which are displayed in the table below.

Track	Artist	Released	Length (m:s)	No. of Streams
Creep	Radiohead	1992	3:59	1271293243
Blank Space	Taylor Swift	2014	3:52	1355959075
Something In The Way	Nirvana	1991	20:37	368646862
Nonsense	Sabrina Carpenter	2022	2:43	342897938
Drivers License	Olivia Rodrigo	2021	4:02	1858144199
Iris	The Goo Goo Dolls	1998	4:50	1284942608

   (a) State which of the five variables contain:

      i. Categorical data. ii. Discrete data. iii. Continuous data.

   (b) If it was suspected that one of the tracks is an outlier in terms of length, state what could be calculated in order to determine whether it is in fact an outlier.

2. A sports clothing store is seeking to better understand its sales of running shoes. For a random sample of twelve sales of pairs of running shoes, the sizes are recorded as follows:

$$4 \quad 7 \quad 9 \quad 11 \quad 6 \quad 9 \quad 8 \quad 4 \quad 10 \quad 9 \quad 6 \quad 5$$

   (a) Contruct a dot plot to display the data.
   (b) Find the median and interquartile range for the sizes of running shoes sold.
   (c) Calculate the mean and standard deviation for the sizes of running shoes sold.
   (d) State the type of data recorded.

3. The back-to-back stem-and-leaf diagram below shows the heights of trees in two areas of woodland.

Woodland A		Woodland B
5	2	1 3 3 8
7 4	3	0 2 4 7
9 7 7 2	4	2 5
7 6 3	5	

   (a) Suggest one improvement needed for the diagram.
   (b) Make one comment comparing the heights of trees in the two areas of woodland.

4. The petal lengths (in centimetres) for samples of flowers from two species of iris flowers is recorded: *iris setosa* and *iris versicolor*. The data is summarised in the table below using statistical software:

species	n	min	Q1	Q2	Q3	max
iris setosa	50	1.00	1.40	1.50	1.58	1.90
iris versicolor	50	3.00	4.00	4.35	4.60	5.10

   (a) For the species *iris setosa*, determine whether the dataset contains any outliers.
   (b) Make two comments comparing the petal lengths of the two species of iris.

2

An Introduction to Probability Theory

The purpose of statistics is to make sense of the world by analysing data. Typically however, it is only possible to obtain data from a random sample of the full population of interest, creating an element of uncertainty that must be navigated carefully. For this reason, developing an understanding of statistical analysis begins with learning about probability theory.

Key definitions

In probability theory, an experiment for which all of the possible *outcomes* can be defined, and yet the actual outcome of any given *trial* cannot be known in advance, is called a *random experiment*. The set of all possible outcomes is called the *sample space* of the random experiment. Probability is assigned to individual outcomes or to sets of outcomes, called *events*. Probability itself is defined as the **long-run relative frequency** of an outcome or event occurring, and values range from 0 (*impossible, and will never happen*) to 1 (*certain, and will happen every time*).

For example, in the context of rolling a fair, cubical die:

- the procedure of rolling the die and observing the result is a **random experiment**.
- each roll of the die may be referred to as a **trial**.
- the **sample space** consists of a list of all of the possible **outcomes**: $\{1, 2, 3, 4, 5, 6\}$
- the **probability** of a 5 is $\frac{1}{6}$ since *in the long-run* it can be expected to occur $\frac{1}{6}$ of the time.
- a possible **event** of interest may be that the outcome is *prime*: $\{2, 3, 5\}$

Probability notation

The probability that a fair cubical die lands on 5 is *one-in-six*, or $\frac{1}{6}$. Using probability notation:

$$\mathsf{P}(5) = \frac{1}{6}$$

2.1 Types of Probability

Experimental Probability

The probability of an event can be estimated through repeated trials - this is referred to as the *experimental probability* of the event. For example, if a motorist passes a set of traffic lights every morning and wishes to know the probability of having to wait at a red light on any given morning, they could record the *proportion* of times they have to wait at a red light over a period of time. If, over the course of 200 mornings, they have to wait at a red light 114 times:

$$\text{P(red light)} \approx \frac{114}{200} = 0.57$$

Theoretical Probability

When each of the outcomes in the sample space of a random experiment can be reasonably assumed to be **equally likely** then the probability of any outcomes or events can be determined. This is called *theoretical probability*.

Equally likely outcomes typically arise from the symmetry of a physical object (such as a coin or die) or a reasonable assumption. The probability of any *individual outcome* in a sample space containing N equally likely outcomes is $\frac{1}{N}$. For example, if a playing card is picked at random from a standard pack of 52 cards, containing no jokers, then the probability of picking the Queen of Hearts is $\frac{1}{52}$.

An **event** is a set of one or more outcomes from the sample space. The probability of an event is given by:

Probability of an Event

$$\text{P(event)} = \frac{\text{number of outcomes from the sample space in the event}}{\text{total number of outcomes in the sample space}}$$

Example

Problem: A bag contains 13 red counters, 5 blue counters and 4 green counters. A counter is chosen at random. Find the probability that the counter is green.

Solution:

$$\text{P(green)} = \frac{4}{22} = \frac{2}{11}$$

In probability theory, *"and"* means that *both* events are true, and *"or"* means that *at least one* is true.

Example

Problem: A car dealership has 80 cars in its showroom. They have a mix of electric, hybrid, petrol and diesel cars, both new and second-hand. The table below shows the numbers of each type.

	electric	hybrid	petrol	diesel
new	15	17	28	3
second-hand	1	7	4	5

A car is selected at random. Determine the probability that the car selected is a second-hand hybrid car.

Solution:

$$\text{P(second-hand and hybrid)} = \frac{7}{80} = 0.0875$$

In this textbook, probabilities will be given as simplified fractions or *rounded to four decimal places* for consistency.

Exercise 2.1

1. A letter of the English alphabet is chosen at random. State the probability that the letter chosen is:

   (a) A vowel. (b) A consonant. (c) In the word *statistics*.

2. A bag contains four blue, five green and six red counters. Find the probability that a randomly chosen counter is:

   (a) Blue. (b) Either green or red. (c) Not red.

3. A box of chocolates contains chocolates of three different flavours. Five are caramel, four are orange and three are strawberry. A chocolate is chosen at random. State the value of:

   (a) P(orange). (b) P(orange and strawberry). (c) P(not caramel)

4. As of 2024, the UK contains 76 cities, of which 55 are in England, 8 in Scotland, 7 in Wales and 6 in Northern Ireland. If a city in the UK is selected at random, find the probability that the chosen city is:

   (a) In Scotland. (b) In Wales or Northern Ireland. (c) Not in England.

5. A board game uses green counters numbered from 1 to 8, blue counters numbered from 1 to 6 and red counters numbered from 1 to 4. A counter is chosen at random. Determine the probability that the counter selected is:

   (a) The blue numbered 5. (b) Numbered 5. (c) Blue or numbered 5.

6. A restaurant grades the fruit contained in a delivery as good or bad. The contents of the previous delivery are shown in the table below.

	Apples	Tangerines	Kiwi
Good	27	19	26
Bad	3	1	4

   Letting G represent that the event the fruit chosen is *good*, and A the event that it is an *apple*, find:

   (a) P(G) (b) P$(G$ and $A)$ (c) P(Not A)

7. A hospital contains a number of doctors, nurses, allied health professionals and support staff, who are each assigned to work on one of Wards 1 to 3. The breakdown of this is shown in the table below.

	Doctor	Nurse	Allied	Support
Ward 1	3	7	5	8
Ward 2	4	10	2	9
Ward 3	2	8	1	5

   If member of staff working at the hospital is chosen at random, find the probability that they are:

   (a) A nurse. (b) Not on Ward 2. (c) On Ward 2 but not a doctor

2.2 Sample Spaces as Lists and Grids

It can help to write out sample spaces. Some sample spaces can be easily described using a simple list, but more organisation may be needed for more complicated sample spaces, such as approaching the list in a structured manner or using a grid. Sample spaces are generally useful only if each of the outcomes is equally likely.

If two coins are tossed, it may be tempting to suggest that the probability of both landing tails is one-in-three, since the experiment can result in either no coins landing tails, only one coin landing tails or two coins landing tails. However a list of the four possible *equally likely* outcomes in the sample space shows that this is not the case:

Not equally likely outcomes: $\{\text{no tails}, \text{one tail}, \textbf{two tails}\}$

Equally likely outcomes: $\{\text{HH}, \text{HT}, \text{TH}, \textbf{TT}\}$

The probability of both coins landing tails is therefore *one-in-four*, or $\text{P}(\text{both tails}) = \frac{1}{4}$.

Example

Problem: Three coins are tossed at the same time. Find the probability that exactly two of the coins land tails.

Solution: *A list of equally likely outcomes will help here. Note there are* $2 \times 2 \times 2 = 8$ *outcomes.*

$$HHH \quad HHT \quad HTH \quad \underbrace{HTT} \quad THH \quad \underbrace{THT} \quad \underbrace{TTH} \quad TTT$$

$$\text{P}(\text{exactly two tails}) = \frac{3}{8}$$

Example

Problem: Two fair dice are rolled. Determine the probability that the sum obtained is at least nine.

Solution: *A grid of outcomes will help here. Note there are* $6 \times 6 = 36$ *outcomes.*

first die

second die

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$$\text{P}(\text{sum} \geqslant 9) = \frac{10}{36} = \frac{5}{18}$$

Exercise 2.2

For most of the following questions, it is recommended to write out a full list of all the possible equally likely outcomes. There may be times when a full list would be excessive in length, in which case careful multiplication can reveal the *number* of outcomes, and only the *relevant* ones may be needed to be listed.

1. A coin is tossed and a cubical, fair die numbered 1 to 6 is rolled. Write down:
   (a) A list of all outcomes.
   (b) P(even and tails)

2. Each morning, a teacher picks at random whether to start the day with tea, coffee or apple juice, and also picks at random which one of museli, granola, toast or porridge to eat for breakfast. By first writing down all the possible ways the teacher may start their day, determine the probability that on any particular morning the teacher has:
   (a) Tea and toast.
   (b) Museli but not coffee.
   (c) Neither porridge nor tea.

3. A coin is tossed three times. Find the probability that it shows:
   (a) Tails twice in a row.
   (b) Tails at least twice.

4. A random number is chosen from 1 to 140, inclusive. By writing down all relevant results, determine the probability of the number chosen having digits that sum to 3.

5. Two fair, cubical dice are rolled. Find the probability that the product of the numbers obtained is less than 5.

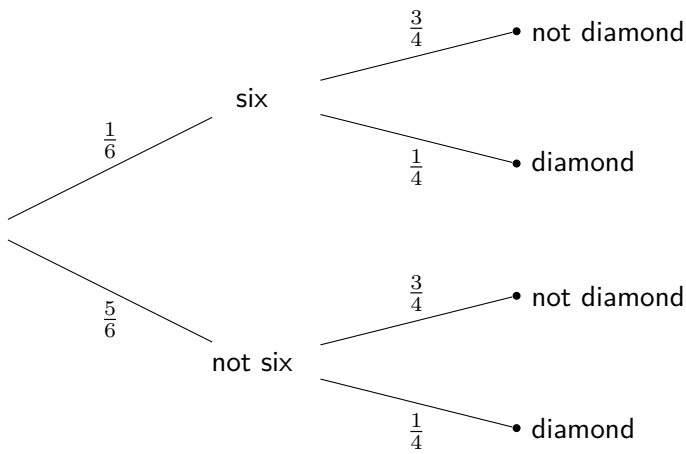
6. A tetrahedral (four-sided) die numbered 1 to 4 and a cubical die numbered 1 to 6 are rolled. Find the probability that at least one of the numbers obtained is 2 or less.

7. Five coins are placed in a bag, with values of 1p, 2p, 5p, 10p and 20p. Two coins are chosen at random, one after another, with the second selected without the first being replaced. Calculate:
   (a) P(the 20p is selected first)
   (b) P(the 20p is selected as one of the two coins)
   (c) P(total of the two coins $\geqslant$ 15p)
   (d) P(total of the two coins $<$ 12p)

8. Two friends, Alex and Charlie, are part of a class which contains eight pupils in total. One pupil from the class is randomly chosen to hand out mini-whiteboards, and different pupil is chosen to hand out pens.
   (a) Determine the number of possible ways the two tasks can be assigned to two pupils.
   (b) Hence, find the probability that:
      i. Alex is chosen to hand out the mini-whiteboards, and Charlie is chosen to hand out the pens.
      ii. Neither Alex nor Charlie are chosen for either task.
      iii. At least one of Alex and Charlie are chosen to perform a task.

2.3 Tree Diagrams

Tree diagrams can be created to map out the possible results of a random experiment, and they are especially useful when considering two or more events. For example, consider a game involving rolling a cubical die and picking a playing card at random from a standard set, with an interest in the chance of the die landing on a *six* and the card picked being a *diamond*:



$$\text{P(six } \textbf{and} \text{ diamond)} = \frac{1}{6} \times \frac{1}{4} = \frac{1}{24}$$

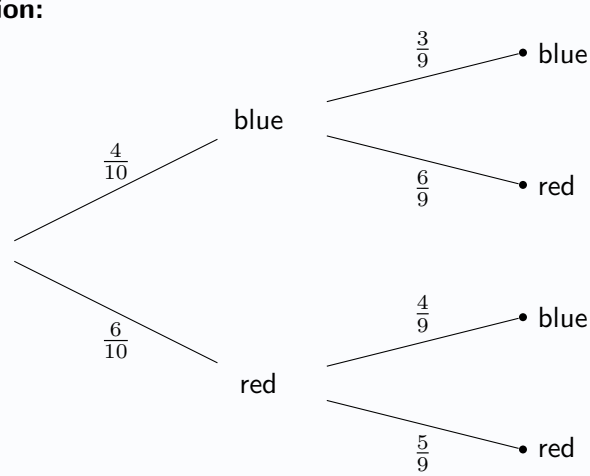
- Every branch must be clearly labelled with a probability.
- The end of each branch must be clearly described, such as *"not diamond"*.
- Sibling branches must add to 1.
- Multiplying the probabilities along branches gives the probability of those events occurring together.

Note that in the tree diagram above the probability of the card being a diamond does not change depending on whether the roll of a die gives a six or not. In the following example the probabilities on the second stage of the tree diagram do change depending on the result of the first stage.

Example

Problem: From a bag containing four blue and six red cubes one is drawn at random, followed by another without replacing the first. Find the probability that two red cubes are drawn.

Solution:



$$\text{P(both red)} = \frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3}$$

Exercise 2.3

Construct a tree diagram for each question.

1. A counter is selected at random from a bag containing four green counters and five yellow counters, and then a second counter is selected from the bag without replacing the first counter that was taken.
   (a) Construct a tree diagram to represent the possible results.
   (b) Find the probability that both counters selected are green.
   (c) Find the probability that at least one of the counters selected is green.

2. A game involves a player randomly selecting one card then a second card from a set of six blue cards and four red cards, with replacement. Calculate the probability of a player selecting two cards of the same colour.

3. In Scotland, 41% of the population have the blood type *O positive (O+)*. A doctor in Scotland randomly selects two of their patients and asks for them to take part in a blood screening trial, as part of which their blood type will be checked. Calculate the probability that neither of the two patients has blood type *O+*.

4. A type of component manufactured for use in wind turbines is tested at two points in the production process, and marked as either *sufficient quality* or *insufficient quality* on the first testing point and then *pass* or *fail* at the second testing point. A component that is not of sufficient quality at the first testing point will continue to be developed and may still pass at the second testing point. Three-quarters of the components are marked as *sufficient quality* initially, of which nine-tenths go on to pass. Of those initially marked as *insufficient quality*, only three-fifths go on to pass.
   (a) Construct a tree diagram to represent this information.
   (b) Determine the probability that a component selected at random passes at the second testing point.

5. A bus for a particular route that is scheduled to run once per day is cancelled on 3% of days. On the days when it is running, it is running late 12% of the time.
   (a) Construct a tree diagram to represent this information.
   (b) Calculate the probability that the bus is running on time on any given day.

6. A pencil case contains pencils of three different types: four HB pencils, two 2B pencils and one 2H pencil. Two pencils are selected at random, without replacement.
   (a) Construct a tree diagram to represent the types of the two pencils selected.
   (b) Find the probability that both pencils selected are of the same type.
   (c) Determine the probability that the two pencils selected are both of a different type.
   (d) Find the probability that neither pencil selected is an HB pencil.

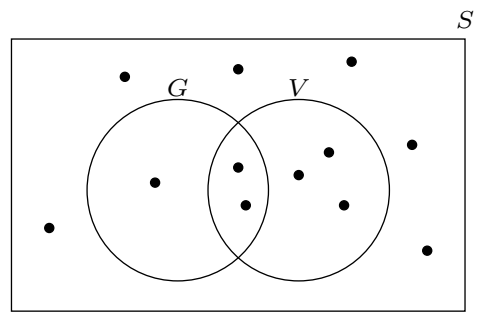
7. A motorist passes through three sets of traffic lights on their way to work each day. If the probability of having to stop at a red light for each set of lights is 0.6, 0.2 and 0.3 respectively then find the probability that:
   (a) The motorist does not have to stop.
   (b) The motorist has to stop at least once.
   (c) The motorist has to stop at exactly two of the sets of lights.

2.4 Venn Diagrams and Set Notation

Venn diagrams are used to show which elements of a set belong or don't belong to one or more subsets. They can offer an alternative way of representing information about a sample space that might otherwise be shown in a table. For instance, the table below shows the twelve dishes from a restaurant's main course menu, broken down by whether or not each is *gluten-free* (G) and whether or not each is *vegan* (V). Beside it is a Venn diagram showing the same information.

	vegan	not vegan
gluten-free	2	1
not gluten-free	3	6

Table



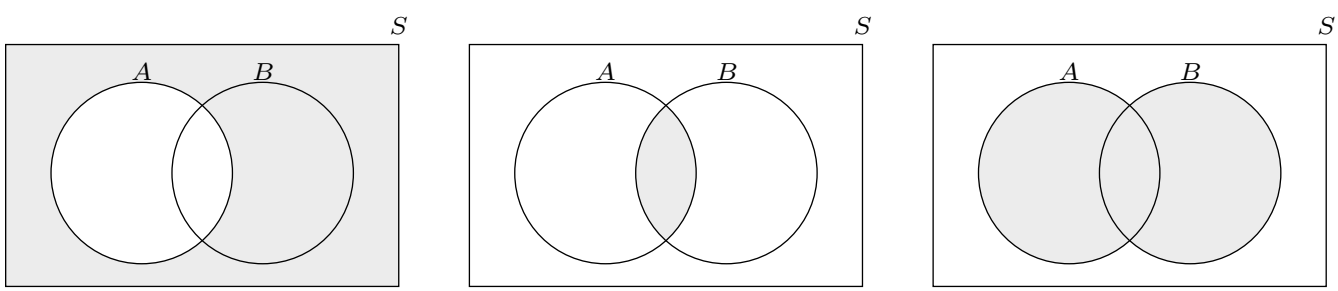
Venn diagram

Complements, Intersections and Unions

So far, *descriptive* words have been used to refer to different sets of outcomes. Using *set notation*:

Description	Set notation	Name
not A	$\overline{A}$ or A'	Complement of A
A *and* B	$A \cap B$	Intersection of A and B
A *or* B	$A \cup B$	Union of A and B

Venn diagrams help illustrate these. Note that *capital letters* are commonly used to represent events.



$\overline{A}$ or A' $\qquad$ $A \cap B$ $\qquad$ $A \cup B$

Example

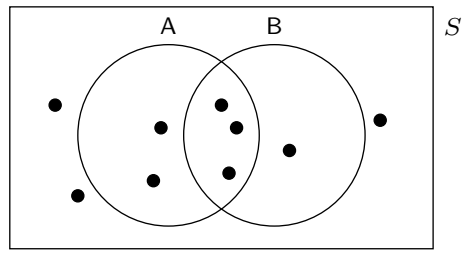
Problem: A dish is picked at random from the twelve on the main course menu shown in the Venn diagram above. State the value of $\mathrm{P}(G \cap V)$.

Solution:
$\mathrm{P}(G \cap V) = \dfrac{2}{12} = \dfrac{1}{6}$

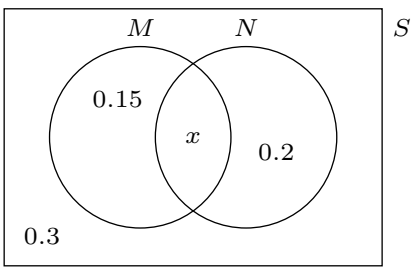
Exercise 2.4

1. One of the nine elements shown on the Venn diagram below is selected at random. State:

   (a) P(A)
   (b) P(B)
   (c) P$(\overline{A})$
   (d) P$(\overline{B})$
   (e) P$(A \cap B)$
   (f) P$(A \cup B)$
   (g) P$(\overline{A} \cap B)$
   (h) P$(A \cap \overline{B})$

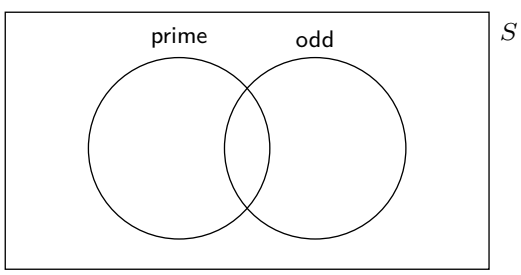


2. The Venn diagram below, containing events M and N, shows the *probabilities* associated with each subset.



   (a) State the value of x.
   (b) State:
      i. P(N).
      ii. P$(\overline{M})$.
      iii. P$(M \cup N)$.

3. Copy the Venn diagram below, writing each of the integers from 1 to 16 inclusive in an appropriate position. Then, given that an integer from 1 to 16 inclusive is selected at random, state the value of:



   (a) P(prime)
   (b) P(odd)
   (c) P($\overline{\text{prime}}$)
   (d) P($\overline{\text{odd}}$)
   (e) P(prime $\cap$ odd)
   (f) P(prime $\cup$ odd)
   (g) P($\overline{\text{prime}}$ $\cap$ odd)
   (h) P(prime $\cap$ $\overline{\text{odd}}$)

4. 100 pupils are asked whether or not they love cats, and whether or not they love dogs, with the results shown in the table below. One pupil is to be selected at random. Let event C represent "a pupil who loves cats is selected", and event D represent "a pupil who loves dogs is selected". By first representing this information on a Venn diagram, find:

   (a) P$(C \cap D)$
   (b) P$(C \cup D)$
   (c) P$(\overline{C} \cap \overline{D})$
   (d) P$(\overline{C} \cup \overline{D})$
   (e) P$(C \cap \overline{D})$
   (f) P$(\overline{C} \cap D)$
   (g) P$(C \cup \overline{D})$
   (h) P$(\overline{C} \cup D)$

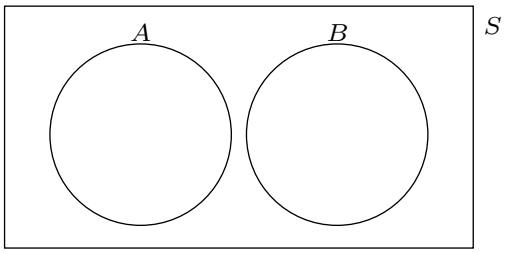
	love cats	do not love cats
love dogs	34	12
do not love dogs	45	9

5. Amongst a class of 24 pupils, seven are studying biology *but not chemistry*, five are studying chemistry *but not biology*, and six are studying neither biology nor chemistry. Find the probability that a pupil chosen at random is studying biology.

6. Of the twenty air pollution monitoring stations in an urban area, nine are showing high levels of particulate matter (PM) in the air and eight are showing high levels of nitrogen dioxide (NO_2) in the air, whilst five are not showing high levels of either pollutant. Determine the probability that one of the twenty stations chosen at random is showing high levels of PM *and* NO_2.

2.5 The Addition Rule for the Union

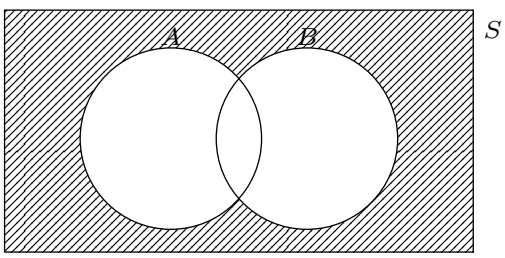
If two sets A and B are such that *no elements lie in both A and B*, then they are said to be **mutually exclusive**. This is sometimes visualised on a digram by two *disjoint* circles, that do not intersect. In terms of probability theory, this means that the probability of the event "A *and* B" is 0 (impossible).



For Mutually Exclusive events A and B:

$$\mathsf{P}(A \cap B) = 0$$

If two sets A and B are such that *every element lies in at least one of A or B*, then they are said to be **exhaustive**. This can be visualised on a Venn digram by crossing out the area of the sample space outside of the two circles. This means that the probability of the event "A *or* B" is 1 (certain).



For Exhaustive events A and B:

$$\mathsf{P}(A \cup B) = 1$$

As well as the two rules above which apply depending on whether A and B are know to be mutually exclusive, exhaustive, or both, the two formulae below apply for *all events* A, B:

The Complement of A:

$$\mathsf{P}(A) + \mathsf{P}(\overline{A}) = 1$$

The Addition Rule for events A and B:

$$\mathsf{P}(A \cup B) = \mathsf{P}(A) + \mathsf{P}(B) - \mathsf{P}(A \cap B)$$

It should be noted that there are generally numerous different ways to approach probability problems, including using a Venn diagram, a tree diagram or formulae. Whilst the problems on the next page have been designed to allow practice using formulae, a flexible approach may help.

Example

Problem: Given events F and G such that $\mathsf{P}(\overline{F}) = 0.3$, $\mathsf{P}(G) = 0.4$ and $\mathsf{P}(F \cap G) = 0.1$ show that F and G are exhaustive.

Solution:

$$\begin{aligned}\mathsf{P}(F \cup G) &= \mathsf{P}(F) + \mathsf{P}(G) - \mathsf{P}(F \cap G)\\ &= 0.7 + 0.4 - 0.1\\ &= 1 \quad \therefore \text{ events } F \text{ and } G \text{ are exhaustive}\end{aligned}$$

Exercise 2.5

1. Given events X and Y such that $\mathsf{P}(X \cap Y) = 0.3$, $\mathsf{P}(X) = 0.4$ and $\mathsf{P}(Y) = 0.5$, find $\mathsf{P}(X \cup Y)$.

2. If $\mathsf{P}(\overline{A}) = \dfrac{1}{5}$, $\mathsf{P}(B) = \dfrac{2}{3}$ and $\mathsf{P}(A \cap B) = \dfrac{1}{2}$, determine the value of $\mathsf{P}(A \cup B)$.

3. Calculate the value of $\mathsf{P}(E \cup F)$ given $\mathsf{P}(E) = 0.32$, $\mathsf{P}(F) = 0.29$ and that events E and F are *mutually exclusive*.

4. Given mutually exclusive events C and D such that $\mathsf{P}(\overline{C}) = \dfrac{5}{7}$ and $\mathsf{P}(D) = \dfrac{1}{2}$, find $\mathsf{P}(C \cup D)$.

5. Calculate the value of $\mathsf{P}(A \cap B)$ if $\mathsf{P}(A) = 0.591$, $\mathsf{P}(B) = 0.472$ and $\mathsf{P}(A \cup B) = 0.9$.

6. If $\mathsf{P}(\overline{X}) = \dfrac{17}{80}$, $\mathsf{P}(\overline{Y}) = \dfrac{29}{40}$ and $\mathsf{P}(X \cup Y) = \dfrac{19}{20}$, determine $\mathsf{P}(X \cap Y)$.

7. Given *exhaustive* events V and W such that $\mathsf{P}(V) = 0.4$ and $\mathsf{P}(\overline{W}) = 0.2$, find $\mathsf{P}(V \cap W)$.

8. Given events H and J such that $\mathsf{P}(H) = \dfrac{3}{8}$, $\mathsf{P}(H \cup J) = \dfrac{7}{9}$ and $\mathsf{P}(H \cap J) = \dfrac{1}{2}$, find $\mathsf{P}(\overline{J})$.

9. If X and Y are mutually exclusive, $\mathsf{P}(X) = 0.3$ and $\mathsf{P}(\overline{Y}) = 0.4$, calculate $\mathsf{P}(X \cup Y)$.

10. Events A and B are exhaustive. If $\mathsf{P}(A) = \dfrac{1}{3}$ and $\mathsf{P}(B \cap A) = \dfrac{1}{6}$, calculate $\mathsf{P}(\overline{B})$.

11. In a fish and chip shop, the probability that a customer's order includes both fish and chips is 0.7, whilst the probability that an order will at least include at least one of fish or chips is 0.95. If the probability that an order contains chips is 0.8, calculate the probability that an order contains fish.

12. If, for exhaustive events M and N, $\mathsf{P}(M \cap N) = \dfrac{1}{4}$ and $\mathsf{P}(M) = \mathsf{P}(N)$, calculate $\mathsf{P}(\overline{N})$.

13. Given events R and Q such that $\mathsf{P}(\overline{Q}) = 0.11$, $\mathsf{P}(R \cap Q) = 0.75$ and $\mathsf{P}(R \cup Q) = 0.95$, find $\mathsf{P}(\overline{R})$.

14. For each pair of events, determine whether they are exhaustive or not:
    (a) X and Y given $\mathsf{P}(X) = 0.4$, $\mathsf{P}(Y) = 0.7$ and $\mathsf{P}(X \cap Y) = 0.2$.
    (b) R and T given $\mathsf{P}(R) = \dfrac{4}{5}$, $\mathsf{P}(\overline{T}) = \dfrac{1}{10}$ and $\mathsf{P}(T \cap R) = \dfrac{7}{10}$.
    (c) E and F given $\mathsf{P}(E) = 0.31$, $\mathsf{P}(F) = 0.82$ and $\mathsf{P}(E \cap F) = 0.28$.

15. For each pair of events, determine whether they are mutually exclusive or not:
    (a) W and V given $\mathsf{P}(W) = 0.35$, $\mathsf{P}(V) = 0.45$ and $\mathsf{P}(W \cup V) = 0.5$.
    (b) X and Y given $\mathsf{P}(X) = \dfrac{1}{3}$, $\mathsf{P}(\overline{Y}) = \dfrac{4}{5}$ and $\mathsf{P}(X \cup Y) = \dfrac{8}{15}$.
    (c) A and B given $\mathsf{P}(\overline{A}) = 0.5$, $\mathsf{P}(\overline{B}) = 0.8$ and $\mathsf{P}(\overline{B} \cap \overline{A}) = 0.3$.

Review Exercise

1. A box contains five red balls numbered from 1 to 5, and four yellow balls numbered from 1 to 4. A ball is selected at random from the box. Let *yellow* be the event "the ball selected is yellow".

   (a) State $\mathsf{P}(\overline{\text{yellow}} \cup 1)$.

   The ball previously selected is returned to the box. Two balls are then randomly taken from the box, without replacement.

   (b) Determine the probability that neither of the two balls taken from the box are yellow.

2. The table below shows the pupil numbers within a school corridor during a particular period, broken down by subject and course level.

	National 5	Higher
English	21	15
History	17	11

   A pupil is selected at random to run an errand. State:

   (a) $\mathsf{P}(\text{English} \cap \text{National 5})$

   (b) $\mathsf{P}(\text{History} \cup \text{Higher})$

3. For the exhaustive events E and F, such that $\mathsf{P}(F \cap E) = 0.7$ and $\mathsf{P}(\overline{E}) = 0.2$, determine $\mathsf{P}(F)$.

4. A parking ticket machine in a remote location in the Cairngorms National Park takes payment by contactless card reader, which connects to the banking network via a 3G phone signal. One some days, the connection fails and the machine cannot take payment. When the humidity is high, the connection will fail on 18% of days. When the humidity is not high, the connection will fail on 7% of days. Humidity is high in the area on 34% of days. Determine the probability on a randomly selected day of the year that the parking machine can take payment.

5. A board game begins with an *defending* player rolling a tetrahedral die with sides numbered from 1 to 4, and a *attacking* player rolling a cubical die with sides numbered from 1 to 6. The player who obtains the highest number earns a point, with a tie leading to the dice being rolled again.

   (a) Find the probability that the the first point is earned, by either player, without the dice having to be rolled again following a tie.

   (b) Find the probability that the attacking player earns the first point.

   If the attacking player wins three points in a row, they are then awarded a *bonus roll* in which they roll *both* dice and gain a bonus point if the product of the numbers obtained is greater than 16.

   (c) Determine the probability that a bonus roll results in a bonus point.

6. The 100 members of an orchestra are shown in the table below, broken down by the section they are in and whether they can attend the upcoming concert.

	String	Woodwind	Brass	Percussion
attend	56	12	10	4
not attend	8	4	5	1

   A member of the orchestra is selected at random. Let B be the event the member selected is in the *brass* section and A that they can *attend* the upcoming concert. State $\mathsf{P}(B \cup \overline{A})$.

7. Exhaustive events A, B and C are such that $\mathsf{P}(A) = 0.25$, $\mathsf{P}(A \cap B) = 0.1$, $\mathsf{P}(\overline{B}) = 0.3$ and $\mathsf{P}(A \cup C) = 0.6$. Events A and C are mutually exclusive. Determine the value of $\mathsf{P}(B \cap C)$.

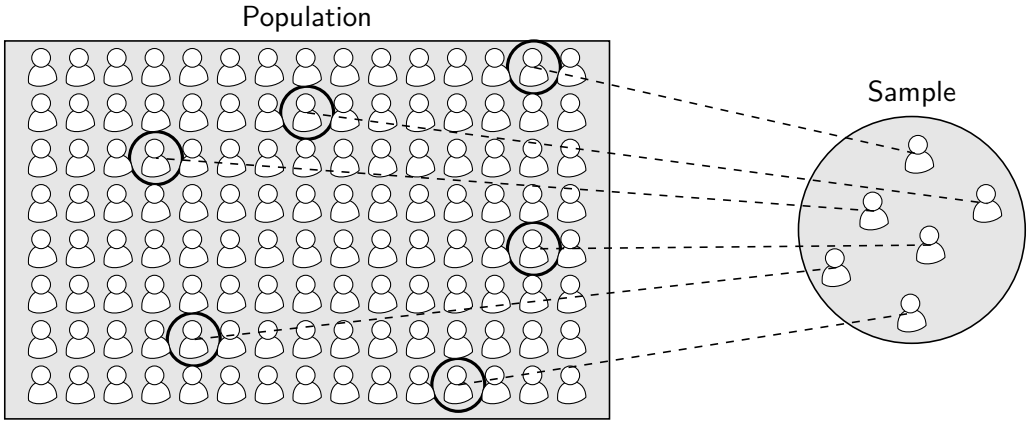
3

Sampling Theory

"Garbage In, Garbage Out" is a phrase used in computer science that may equally be applied to statistics. It should be discouraged to think that the task of a statistician begins with raw data; the manner of the collection of the data itself is of fundamental importance. Any statistical inferences made on the basis of data that has not be collected with an appropriate *sampling method* should not be trusted.

Populations and Samples

To recap from chapter 1, the term **population** refers to the entire set of objects in which we are interested. A **sample** is a selection of some objects chosen from the population.



A **census** is used to collect information from a full population and a **sample survey** is used to collect data on a sample.

It is usually very difficult to carry out a census as it is time consuming, expensive, and requires an accurate and complete list of every member of the population. In some cases, investigating an entire population would destroy it all, such as the burn time of candles. It is therefore usually preferable that a sample survey is carried out and, if representative of the population, it should give an accurate indication of the characteristics of the population.

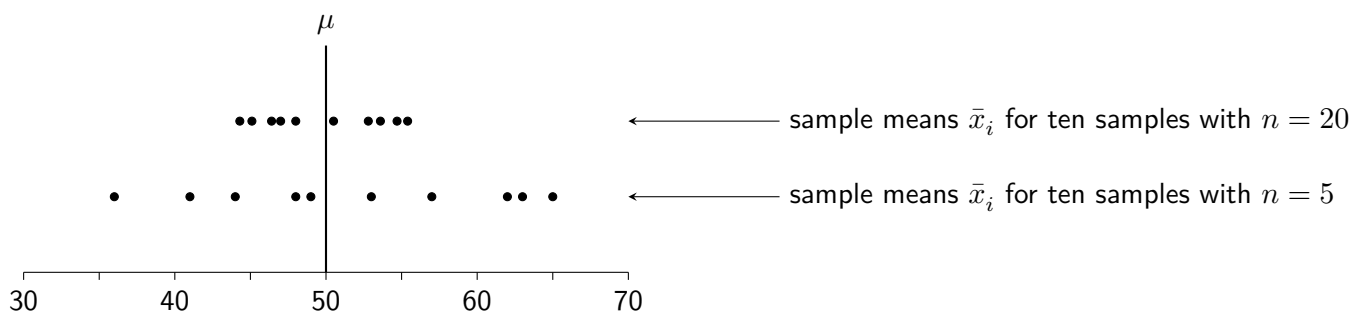
3.1 Sampling Considerations

Samples are used to infer information about the populations from which they are taken, and sample data collected using higher quality sampling techniques should tend to lead to more reliable inferences. There are several key considerations and pitfalls to be aware of whenever the collection of sample data is being planned.

Sample Size

Parameters, such as the *population mean*, are generally considered as fixed, but usually unknown values. If samples are repeatedly taken from a population and their means are are obtained, the values of those sample means will sometimes be higher than that of the true population mean and sometimes lower. This comes from *random sampling error* which, despite its name, is a natural effect due to chance variation.

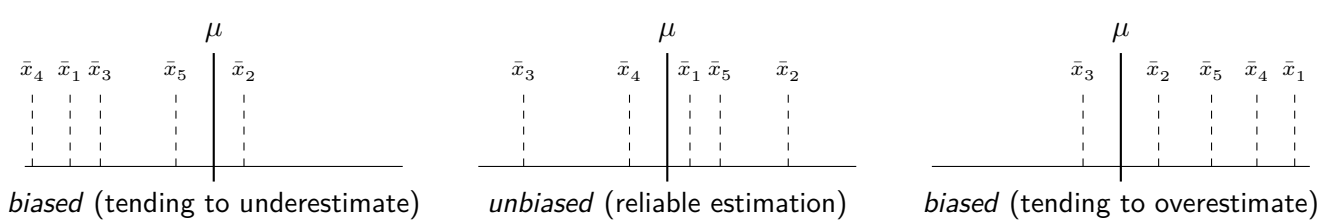
The figure below shows two sets of sample means for random samples taken from the same distribution with a population mean of $\mu = 50$. For one set a sample size of $n = 5$ was used, with $n = 20$ for the other.



It can be seen that larger sample sizes tend to produce sample statistics that better represent the population parameters they are used to estimate. Larger samples are therefore generally preferred, and statistical studies conducting using only small samples of data should typically be treated with more caution than those based on larger samples. When small samples are used, the reasons are usually one or more of cost, time, and availability of data.

Bias

Poor sampling technique may lead to samples that are not *representative* of the population from which they are drawn. The most common effect of this is that *bias* is introduced: a tendency for sample statistics calculated from these samples to overestimate, or underestimate, the population parameter they aim to represent. The diagrams below illustrate the effects various sampling techniques may have on sample means obtained from repeated sampling; the techniques used for the left and right diagrams are causing bias, whilst the sample means obtained for the middle diagram are *unbiased*.

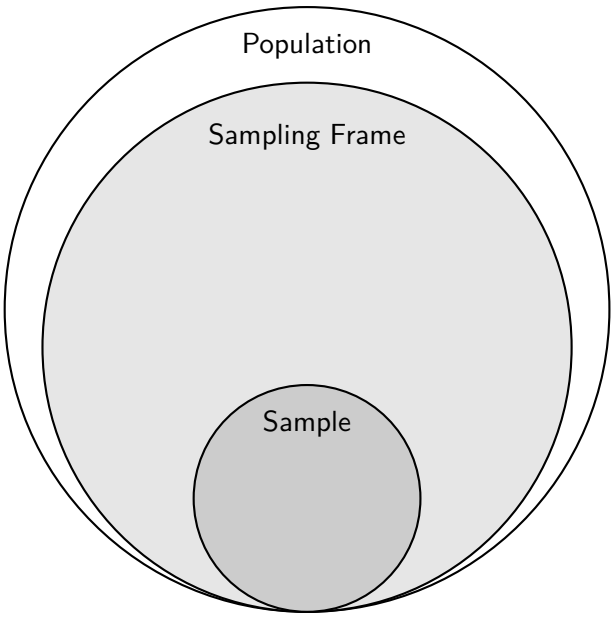


It is very difficult to identify that bias has been introduced, and effectively impossible to correct it. The only reliable way to avoid introducing bias is through application of *appropriate random sampling techniques*.

Sampling Frames

Any real-life experience of sampling for any kind of study is likely to quickly reveal that it is not quite as straightforward as *"just choosing at random"*. Several key elements must be identified and problems considered.

For example, consider the context of wishing to use registration time one morning to survey a sample of pupils at a large secondary school for their views on the school cafeteria.



The **target population** is the *group being studied*.

e.g. all of the pupils who attend a school

The **sampling frame** is the part of the target population, of size N, from which a sample can be taken, typically in the form of an ordered list.

e.g. a numbered list of the school roll

The **sample** consists of the subset of the sampling frame selected to be studied, with size n.

e.g. those pupils selected to be surveyed

Sampling units, which make up the sampling frame, may be individual people, buildings, measurements taken at different times, square grids on a map, and so on.

The *sampling method* is the way in which sampling units are selected from the sampling frame to create the sample.

Care should be taken to ensure that sampling frames are representative of the target population. For example, if walking habits are being studied, a sampling frame containing only residents of Glasgow city centre is likely to be *not representative* of an intended target population of people living in Scotland.

The impact to a study of an ill-defined target population, an unrepresentative samplng frame or a poor sampling technique is likely to be data which is *biased* and *not representative* of the population, and *unreliable conclusions*.

Exercise 3.1

1. A researcher wants to study the shopping habits of those who use his local supermarket. To collect data, he plans to stand at the exit of the self-checkout area one Sunday morning and ask shoppers if he can take a photo of their receipt. Since he will not have time to ask everyone, he will pick only those he feels are most likely to agree to this.
   (a) Identify:
      i. The target population.
      ii. The sampling frame.
   (b) Suggest two problems with the sampling frame chosen, and explain the impact this may have on their data.
   (c) Comment on the suitability of the sampling method the researcher plans to use.
2. A large field is full of many wildflowers. A botanist wishes to conduct a survey to assess the health and variety of species of the wildflowers. A random number generator will be used to help obtain a sample.
   (a) Identify one problem with a proposal to use individual flowers as sampling units, to make up a numbered sampling frame from which to draw a sample.
   (b) Suggest an alternative sampling unit for the botanist to use instead.
   (c) Explain the purpose of a random number generator in producing a sample.

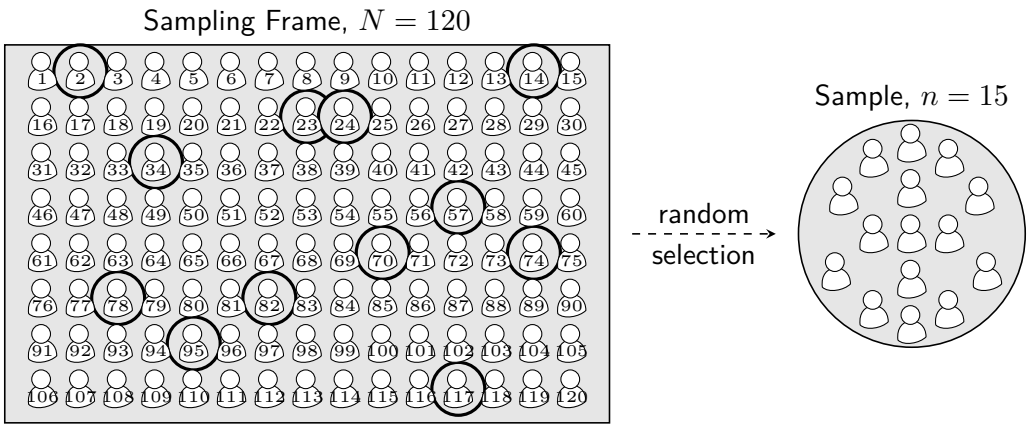
3.2 Six Sampling Methods

In this course, knowledge is required of *four random samplng methods* and *two non-random sampling methods*.

Simple Random Sampling (Random)

With *simple random sampling* each unit of the sampling frame is *equally likely* to be selected, such as by drawing names out of a hat. In general, a simple random sample of size n can be selected from a population of size N by numbering the units in the sampling frame from 1 to N, then selecting n unique random numbers using a random number generator. Those units in the sampling frame corresponding to those numbers generated are then selected to form the sample.

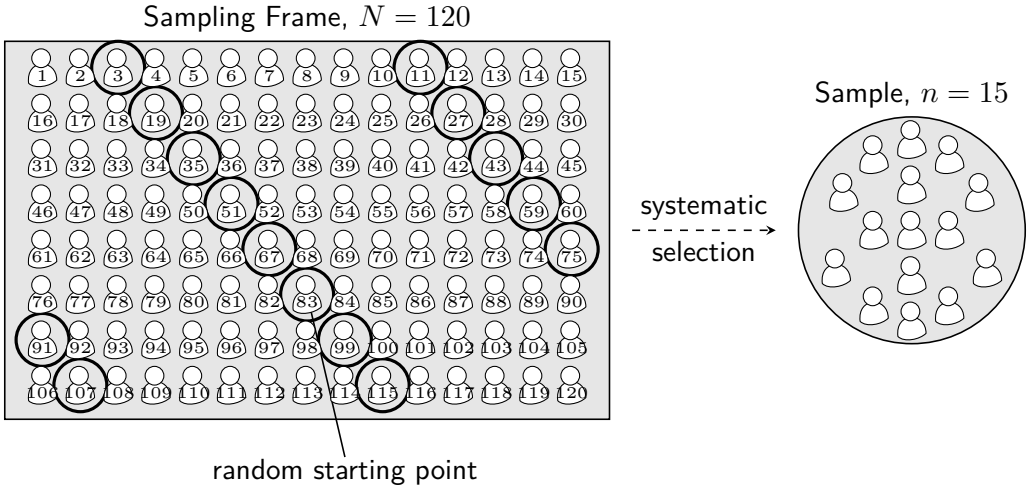
For example, simple random sample of 15 pupils from a year group of 120 can be made by numbering a list of pupils from the year group from 1 to 120, then selecting 15 random numbers using a random number generator.



Systematic Sampling (Random)

Systematic sampling involves using an orderly *pattern* to select units, starting from a randomly selected point. In general, for a numbered sampling frame of size N, a random number generator may select a number from 1 to N, with the corresponding sampling unit selected. After that initial selection, *every subsequent* $\frac{N}{n}th$ *unit is selected*, looping back to the start of the sampling frame as necessary to complete the sample of size n.

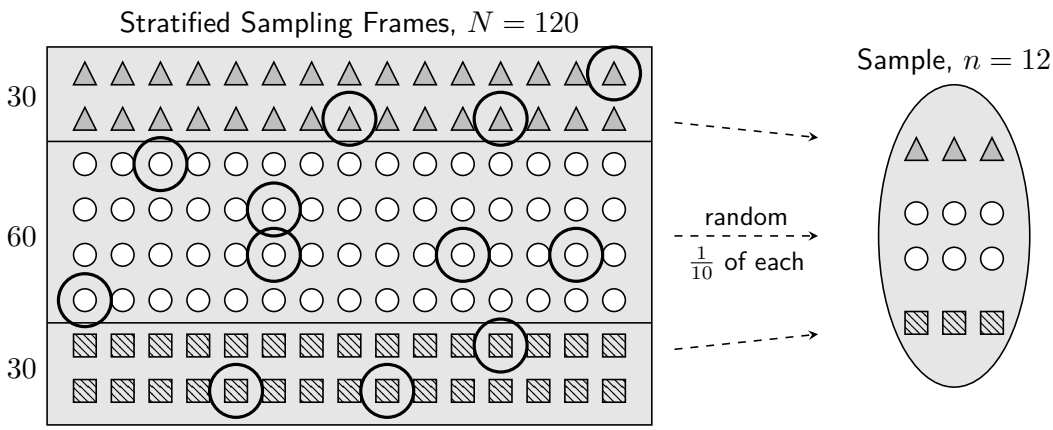
For example, a systematic sample of 15 pupils from a year group of 120 can be made by first selecting a random starting point, such as the $83rd$ pupil, in an alphabetised list of pupils in the year group. From there, every $\frac{120}{15} = 8th$ pupil from the list is chosen, looping back round at the end of the list until 15 pupils have been chosen.



Stratified Sampling (Random)

If a target population contains distinct *strata*, where each stratum may have its own distinct characteristics, *stratified sampling* ensures that each strata will be *proportionally represented*. In general, this involves obtaining an ordered sampling frame for *each stratum* and using the proportions of each stratum in the population to determine how many to sample from each, using simple random sampling. Finally, these samples from the strata are combined.

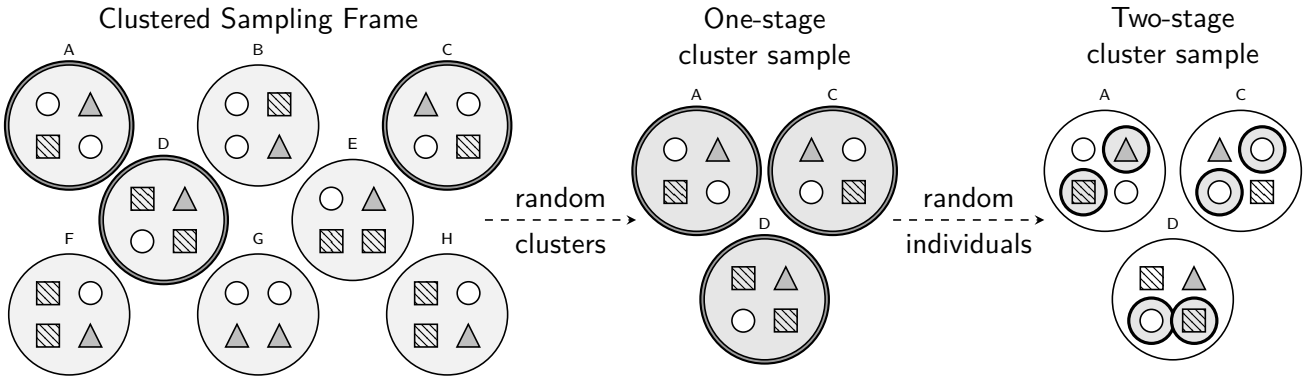
For example, a hospital wishing to survey its staff about the food in the staff canteen may split the $N = 120$ staff into management, doctors/nurses and support staff. For a sample of size $n = 12$, covering $\frac{1}{10}th$ of the sampling frame, simple random sampling is used to select $\frac{1}{10}$ of the staff *from each group*. These are then collected to form the stratified sample.



Cluster Sampling (Random)

If a target population is split into clusters, and each cluster represents a similar mix of characteristics to the full population, *cluster sampling* may be useful. For a *one-stage* cluster sample, one or more clusters are randomly selected, from a sampling frame of clusters, and then *all individual* within those clusters form the sample. For a *two-stage* cluster sample, an additional level of sampling takes place to only sample *some* of the individuals within each cluster.

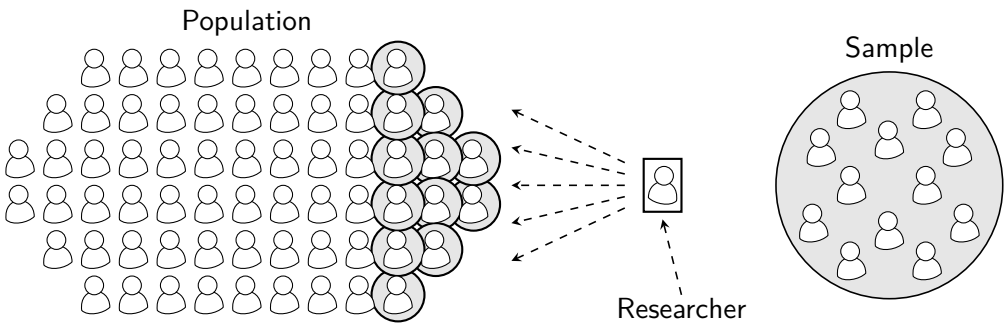
For example, a hospital wishing to survey its staff about the food in the staff canteen may recognise that staff are split into hospital wards A to H. Several wards are randomly selected (A, C and D) and each member of staff working there surveyed. For a two-stage sampling method, simple random sampling could be used to select only *some* of the ward staff from the selected clusters to sample rather than everyone.



The following sampling methods are *non-random*. Whilst they may seem helpful in some instances for quickly gathering data, their lack of random process may lead to *bias*, and any inferences made using such sampling methods may be considered unreliable.

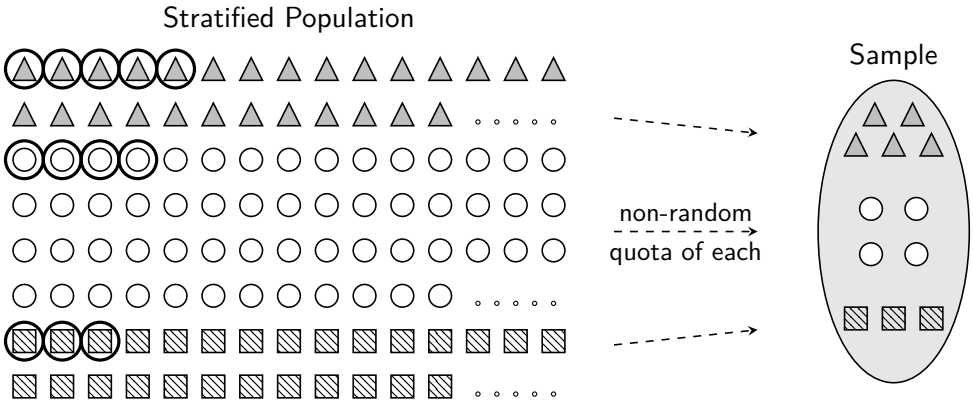
Convenience Sampling (Non-random)

Taking a sample of a target population taken from an easily obtainable group, with no random process taking place, is called *convenience sampling*. For example, a teacher wishes to investigate the heights of pupils in a particular year group and uses pupils within their class in that year group as the sample.



Quota Sampling (Non-random)

Where a population contains distinct strata, *quota sampling* sets a target number of individuals from within each stratum to sample. It differs from stratified sampling in that the number sampled from each stratum is according to a set *quota* rather than proportional to their size, and the selection is often *non-random*. This means that ordered *sampling frames are not required* for each stratum, and the proportion that each makes up within the population may not be known.



For example, a council wishing to know local opinion of proposed changes to a park may see the population as split into strata including pensioners, teenagers, parents of pre-school children, and so on. Interviewer are placed on the town's main shopping street and speak to people who pass, choosing for themselves who to interview from their allocated strata.

Advantages and Disadvantages

The possible advantages and disadvantages of sampling methods depend on the context in which they are used. However, desired *advantages* of a sampling method may include it being: cheap to administer; quick to administer; free from bias due to the use of a random process; representative of the full population; easy to obtain a suitable sampling frame for. Conversely, *disadvantages* may include it being: expensive; time-consuming; non-random, possibly introducing bias; not representative of the full population; difficult to obtain a suitable sampling frame for.

The method of collecting data may also impact the quality of the data obtained. For example, *emailing* selected individuals a questionnaire may introduce bias by excluding those who don't use email and allowing *self-selection*.

Exercise 3.2

1. For each, state the sampling method that is being described.
   (a) The population is made up of groups, each of which contains broadly similar characteristics to each other, and the population. One or more of these groups are randomly selected, and everyone in those selected groups are chosen to make up the sample.
   (b) It is recognised that the population contains a number of subgroups, each of which may have their own distinct characteristics. Individuals are randomly selected from each subgroup, such that the proportions of each subgroup in the sample match those proportions of the population.
   (c) Once a first random individual from the popualtion has been selected, subsequent individuals are selected at regular intervals until a sample has been obtained.
   (d) Individuals are selected from the population at random, with each having an equal chance of being selected.
   (e) It is recognised that the population contains a number of subgroups, each of which may have their own distinct characteristics. Ten individuals from each subgroup are selected, without the use of a random process.
   (f) The individuals most readily available are selected.

2. The operators of a cruise ship wish to survey the condition of the fittings in the cabins on board by carefully examining a sample of rooms. Of the 360 cabins, 24 are classed as *Deluxe*, 72 are *Premuim* and the remaining are *Standard*.
   (a) Explain why the ship's operators might wish to use stratified sampling for their survey.
   (b) Describe the steps involved in sampling 30 of the ship's cabins.

3. A marine biologist is interested in the health of the coral reefs on the West Coast of Scotland. On his next scuba dive near the coastal town in which he lives, he takes some time to photograph the animals and plants on and around the coral reef surface. Based on the survey, he concludes that the health of coral reefs on the West Coast of Scotland has deteriorated.
   (a) State the type of sampling method used by the marine biologist.
   (b) Suggest one reason why the conclusion he reached may not be reliable.

4. The owners of a nationwide chain of petrol stations wish to measure the accuracy of fuel delivery by its pumps. They own 126 petrol stations, each containing typically 8 pumps, and they wish to sample approximately 40 pumps.
   (a) State one advantage of using cluster sampling for the survey.
   (b) Describe the steps involved in creating a one-stage cluster sample of 40 pumps.

5. A music lover has a collection of vinyl records, arranged along a single shelf across one wall. Concerned that some of the actual records may be missing from their sleeves, she checks the one on the far left of the shelf and then every 10th record until she reaches the end of the shelf.
   (a) State the sampling method which most closely matches the method used.
   (b) Identify one way in which the steps taken do not match the ideal process of the sampling method stated in part (a).

6. A school wishes to survey the views of pupils, parents and staff on their opinions of a proposed new school emblem design. Wanting to ensure each group is included, during the annual Sports Day questionnaires are given to twelve of the pupils and twelve of the parents standing in the queue to register, and to the eight members of staff helping run the event.
   (a) State the sampling method used.
   (b) State one disadvantage of this sampling method and describe the impact this may have on the data obtained.

Review Exercise

1. The owners of a chain of nine restaurants, located in various cities around Scotland, wish to interview a sample of the employees working in those restaurants.

   (a) One of the owners suggests that he asks to interview any employees that he sees when he dines at one of the restaurants next week.

      i. State the type of sampling method that he is suggesting the owners use.

      ii. State one disadvantage of this sampling method.

   (b) Another owner proposes instead that they could type the name of each of the 150 employees into a spreadsheet and use a random cell selection tool to pick those to be interviewed.

      i. State the type of sampling method that she is proposing.

      ii. Describe how the owners could instead use systematic sampling to select 30 employees.

   (c) It is pointed out by the HR Director that the employees can be broken down into three categories, each of which may have their own distinct views on what it is like to work in the restaurant chain:

   - Managers and supervisors.
   - Kitchen staff.
   - Waiting and bar staff.

   Each of the restaurants contains a similar mix of the different types of staff.

      i. Describe how a one-stage cluster sample could be conducted.

      ii. Suggest one advantage and one disadvantage of using this method, compared to using systematic sampling.

2. A company employs the following personnel:

Management	Sales staff	Shop floor staff
100	400	4500

   Explain how a stratified sample of 150 employees could be obtained, ensuring that each type of employee is proportionally represented.

3. A nutritionist is studying the calorific contents of pizzas sold in the UK, having read that the salt content is much higher than it is in Italy. They obtain a list of the takeaways which sell pizza in their local area by searching on social media, and randomly select twelve for their study. Suggest two reasons why the sampling frame may not be sufficiently representative of the target population.

4. A researcher wishes to interview a random sample of pupils taking Higher Mathematics to investigate the proportion of them that are interested in pursuing a career in a mathematical field.

   (a) Suggest a reason why taking a simple random sample of all pupils taking Higher Mathematics in Scotland would not be feasible.

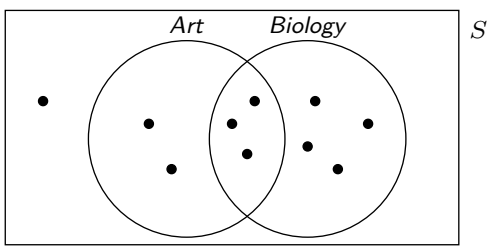
   (b) Explain how a two-stage cluster sample could be taken.

5. A wildflower meadow contains many different species of wildflowers growing together. A botanist wishes to assess the health of the wildflowers in the meadow by studying a sample of the flowers. They have access to a map of the meadow, broken down into boxes with grid references such as A7, and B2. Explain how the botanist could use simple random sampling to obtain a sample.

4

Further Probability Theory

Suppose ten pupils in a class are asked whether they are taking Art, and whether they are taking Biology, and the results are displayed in both a Venn diagram and a table.



	Biology	$\overline{\text{Biology}}$
Art	3	2
$\overline{\text{Art}}$	4	1

From both the table and the Venn diagram, it can be seen that, of the ten pupils, five are taking Art. Therefore, the probability of randomly selecting a pupil from the class and that pupil turning out to be taking Art can be stated:

$$\mathsf{P}(\text{Art}) = \frac{5}{10} = \frac{1}{2}$$

Now suppose a pupil is chosen at random and, before it is known whether or not they are taking Art, they reveal that they are taking Biology. The probability that the pupil is also taking Art is no longer $\frac{1}{2}$, since only three of the seven pupils taking Biology are taking Art. The probability now being sought is the probability that they are taking Art *given that they are taking Biology*. This is called a *conditional probability*, and can be notated as:

$$\mathsf{P}(\text{Art} \mid \text{Biology}) = \frac{3}{7}$$

Conditional probability is fundamental to many of the statistical analysis techniques that will be introduced.

4.1 Conditional Probability from Tables and Diagrams

Example

Problem: A medical trial for drug treating a minor illness has patients given either the drug or a placebo, with a test after one month of treatment determining whether or not each patient has fully recovered.

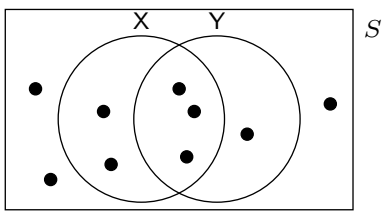
	drug	placebo
fully recovered	42	18
not fully recovered	8	12

A patient is selected at random to be interviewed. State P(fully recovered | placebo).

Solution: P(fully recovered | placebo) $= \dfrac{18}{30} = \dfrac{3}{5}$

Exercise 4.1

1. The Venn diagram below shows equally likely outcomes in a sample space and events X and Y. State:



(a) P(X)
(b) P$(X|Y)$
(c) P$(X \cap Y)$
(d) P$(X \cup Y)$
(e) P(Y)
(f) P$(Y|X)$
(g) P$(Y \cap X)$
(h) P$(Y \cup X)$

2. A basketball coach records the number of 2-pointer, 3-pointer and free-throw scoring attempts over the course of a season, broken down by whether those attempts scored or missed. An attempt is chosen at random to be analysed. Find:

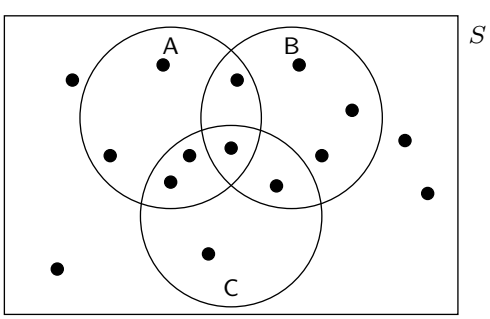
(a) P(scored)
(b) P(scored | 3-pointer)
(c) P(3-pointer)
(d) P(3-pointer | scored)

	2-pointer	3-pointer	free throw
scored	72	30	44
missed	60	52	12

3. On mornings when it is raining, the probability that a red squirrel will leave its drey is 0.4. On mornings when it is not raining, the probability that it will leave its drey is 0.9. It is rainy on 30% of mornings. State the value of:

(a) P(rainy)
(b) P(not rainy)
(c) P(leaves | rainy)
(d) P(does not leave | rainy)
(e) P(leaves | not rainy)
(f) P(does not leave | not rainy)

4. The Venn diagram below shows equally likely outcomes in a sample space and events A, B and C. Determine:



(a) P(A)
(b) P(B)
(c) P(C)
(d) P$(A \cap B)$
(e) P$(B \cup C)$
(f) P$(A \cap B \cap C)$
(g) P$(A|C)$
(h) P$(C|B)$
(i) P$(\overline{B}|C)$
(j) P$(C|\overline{A})$
(k) P$((A \cap B)|C)$

4.2 The Conditional Probability Formula

Where a conditional probability cannot be intuitively stated, the formula for conditional probability may be used:

Conditional Probability:

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$$

It may be helpful to note that $\mathrm{P}(\overline{A}|B) = 1 - \mathrm{P}(A|B)$.

Example

Problem: Given events V and W such that $\mathrm{P}(V) = 0.7236$ and $\mathrm{P}(V \cap W) = 0.1046$, find $\mathrm{P}(W|V)$.

Solution: $\mathrm{P}(W|V) = \dfrac{\mathrm{P}(W \cap V)}{\mathrm{P}(V)} = \dfrac{0.1046}{0.7236} = 0.1446$

Exercise 4.2

1. Find, by first writing out the conditional probability formula for these events:
   (a) $\mathrm{P}(A|B)$ given $\mathrm{P}(B) = 0.8$ and $\mathrm{P}(A \cap B) = 0.16$.
   (b) $\mathrm{P}(\text{red} \mid \text{blue})$ given $\mathrm{P}(\text{red} \cap \text{blue}) = 0.1526$, $\mathrm{P}(\text{blue}) = 0.3147$ and $\mathrm{P}(\text{red}) = 0.2142$.
   (c) $\mathrm{P}(X|Y)$ given $\mathrm{P}(Y) = \frac{1}{2}$ and $\mathrm{P}(Y \cap X) = \frac{1}{3}$.
   (d) $\mathrm{P}(M|N)$ given $\mathrm{P}(\overline{N}) = 0.6137$, $\mathrm{P}(N \cap M) = 0.1249$ and $\mathrm{P}(M \cup N) = 0.4511$.
   (e) Using (a) to (d), state:

      i. $\mathrm{P}(\overline{A}|B)$ ii. $\mathrm{P}(\overline{\text{red}} \mid \text{blue})$ iii. $\mathrm{P}(\overline{X}|Y)$ iv. $\mathrm{P}(\overline{M}|N)$

2. Given events V and W such that $\mathrm{P}(V \cap W) = 0.28$ and $\mathrm{P}(W) = 0.4$, find $\mathrm{P}(\overline{V}|W)$.

3. Numbered cards of different colours are selected at random from a pack during a game such that the probability of a selected card being *yellow* is 20%, and the probability of it being *yellow and a five* is 4%. Find the probability that a selected card is a five *given that it is a yellow*.

4. A football team knows from looking at their past results that they concede at least two goals in $\frac{2}{5}$ of all games. In $\frac{1}{4}$ of all of their games they concede at least two goals and lose the game. Find the probability that, given that they concede at least two goals in a game, they lose.

5. Given events M and N such that $\mathrm{P}(M) = 0.6$, $\mathrm{P}(N) = 0.4$ and $\mathrm{P}(M \cup N) = 0.9$, find $\mathrm{P}(M|N)$.

6. For events V and W, $\mathrm{P}(V) = 0.6$, $\mathrm{P}(V|W) = 0.8$ and $\mathrm{P}(V \cap W) = 0.5$. Calculate $\mathrm{P}(V \cup W)$.

4.3 Independent Events

The probability of tossing a coin and it landing tails is $\frac{1}{2}$. The probability of it landing tails *given that it is a weekday*, notated as P(tails | weekday), is *still* $\frac{1}{2}$. Since it being a weekday or not makes no different to the probability of a coin showing tails, the two events are said to be *independent*.

The probability a randomly selected card from a standard pack of playing cards being a Diamond is $\frac{1}{4}$. The probability the card is a Diamond *given that we discover the card picked is red*, notated as P(Diamond | red), is $\frac{1}{2}$. Since knowing the card picked is red changes the probability of it being a Diamond, the two events are said to be *not independent*.

For Independent A and B:

$$\mathrm{P}(A|B) = \mathrm{P}(A)$$

For Not Independent A and B:

$$\mathrm{P}(A|B) \neq \mathrm{P}(A)$$

Example

Problem: Given $\mathrm{P}(D) = 0.25$, $\mathrm{P}(E) = 0.6$ and $\mathrm{P}(D \cap E) = 0.15$, show that E and D are independent.

Solution:

$$\mathrm{P}(D|E) = \frac{\mathrm{P}(D \cap E)}{\mathrm{P}(E)} = \frac{0.15}{0.6} = 0.25$$

Since $\mathrm{P}(D|E) = \mathrm{P}(D)$, events D and E are independent.

Exercise 4.3

1. For each pair of events X and Y, state whether you would expect them to be independent or not.

   (a) A person is selected at random in a survey:
   - X: They were born in January.
   - Y: They work in an office.

   (b) A mechanic is inspecting a car's tyre treads:
   - X: The front tyres have low tread.
   - Y: The rear tyres have low tread.

2. For each pair of events, determine whether or not they are independent.

   (a) A and B given:
   - $\mathrm{P}(A) = 0.3$
   - $\mathrm{P}(B) = 0.8$
   - $\mathrm{P}(A \cap B) = 0.16$

   (b) X and Y given:
   - $\mathrm{P}(X) = \frac{3}{4}$
   - $\mathrm{P}(Y) = \frac{7}{12}$
   - $\mathrm{P}(X \cap Y) = \frac{7}{16}$

   (c) M and N given:
   - $\mathrm{P}(M \cup N) = 0.67$
   - $\mathrm{P}(\overline{M}) = 0.55$
   - $\mathrm{P}(M \cap N) = 0.18$

3. A music tutor taught 40 students last year, each learning one of violin, trumpet or saxophone, and each have lessons either midweek or at the weekend. A student is chosen at random to take a survey to give feedback. Let V represent that the student chosen took violin lessons, T that they took trumpet lessons and W that they had lessons at the weekend.

	violin	trumpet	saxophone
midweek	12	11	9
weekend	3	4	1

   (a) Show that V and W are independent.

   (b) Show that T and W are not independent.